Ethiopian Association of Anesthetists

Evidence Based Clinical Decision Making

Participants' Manual

February 2023

Bishoftu Ethiopia

Document Management Control:

Authorized by: EAA Educational Advisor

Prepared by: EAA Educational Advisor, Health Facilities and Higher Education Institutions

Approved by: Debre Tabor University CPD center

Date issued: June 2023

Last review: June 2023

Next review: June 2026

Table of Contents

List of Tables and figures III

Acknowledgement IV

Abbreviation V

Introduction to the Manual 1

Rationale 2

Target Audiences 2

Course syllabus 3

Chapter One: Introduction for EBM and Formulating Answerable Clinical Questions 1

1.1: Introduction to Evidence Based Clinical Decision Making (EBCDM) 1

1.1.1 Definition 2

1.1.2 Benefits of EBM 3

1.1.3 Standards and criterias for EBM 3

Summary 7

1. 2: Formulating an Answerable Clinical Question 7

1.2.1 Introduction 8

1.2.2 How to ask well formulated answerable questions? 8

1.2.3 The type of question being asked 10

1. 2.4. The type of study that best answers the question 10

Summary 13

Chapter Two: Evidence retrieval 14

2.1: Study Design and Strength of Evidence for Relevant Clinical Question 14

2.1.1 types of study design 15

2.1.2 Identify types of study design for clinical question 18

Summary 20

2.2 Effective Search Strategy 20

Summary 22

2.3: Sources of evidence 22

2.3.1 The Haynes 5S knowledge acquisition pyramid 23
2.3.2 Filtered and unfiltered sources of evidence 25
Summary 27
2.4 Effective Database search 27
2.4.1 Effective database search 28
Summary 32
Chapter Three: Appraising the Evidence 35
3.1 Critical Appraisal 35
3.2: Critical Appraisal Checklist 38
3.2.1 Introduction 38
3.2.2 Critical Appraisal of Randomized Controlled Trials 39
3.2.3 Critical Appraisal of systematic reviews 41
Chapter Four: Applying Evidences to Practice 44
4.1 Clinical Pathways and Decision analysis 45
4.1.1 Clinical Pathways 45
4.1.2 Decision Analysis 47
Summary 48
4. 2 Assessing Recommendations and Facilitating Implementation of Evidence Based Practice 49
4.2.1 Assessing Recommendations 49
4.2.2 Fascilitator and Barriers to the Implementation of Evidence into Practice 53
Chapter Five: Assess Outcomes, Make Changes to Practice, and Quality Assurance 55
5.1 Assess Outcomes and Make Changes to Practice 56
5.2 Quality Assurance 58
5.2.1 Quality indicators 59
5.2.1 Quality indicator assessment 60
Annexes 63
Annex 1: EBDM Implementation Checklist 63
Annex 2: Selected Articles for Clinical Appraisal 65

List of Tables and figures

List of Tables

Table 1: Chapter/session outline 4
Table 2: Training Schedule 7
Table 3: Types of Common Clinical Questions in Evidence Based Medicine 10
Table 4: Preferred study designs for different study questions 10
Table 5: Types of Common Clinical Questions in Evidence Based Medicine 20
Table 6: Types of Bias 37
Table 7: Critical Appraisal of Randomized Controlled Trials 39
Table 8: Critical Appraisal of Systematic Reviews 41
Table 9: Facilitators and Barriers to the Implementation of EBP 54
Table 10: EBP Self-Assessment Checklist 58

List of Figures

Figure 1: Steps of Evidence based Medicine 5
Figure 2: Algorism of Study Designs 19
Figure 3: The Haynes knowledge acquisition pyramid 23
Figure 4: Sample Decision tree for a patient with hypothetical tumour. 48
Figure 5: Models for Implementation of Evidence based practice 53
Figure 6: Mechanisms of disseminating the results of an EBP intervention 57

Acknowledgement

This CPD program/manual was made possible by the generous support of the American people through the United States Agency for International Development funded Health Workforce Improvement Program (HWIP) under the terms of the Cooperative Agreement No. 72066320CA00008. The contents are the responsibility of the Ethiopian Association of Anesthetists and do not necessarily reflect the views of USAID or the United States Government.

Abbreviation

EAA	Ethiopian Association of Anesthetists
AHA	American Heart Association
EBCDM	Evidence Based clinical Decision Making
EBDM	Evidence Based Decision Making
EBM	Evidence Based Medicine
EBP	Evidence Based Practice
GRADE	Grading of Recommendations Assessment, Development and Evaluation
QI	Quality Indicators
RCT	Randomized Controlled Trial
USPSTF	US Preventive Services Task Force

Introduction to the Manual

This Evidence Based decision making (EBDM) training is 5 days training designed to prepare health professionals to critically appraise and implement available evidences for clinical decision making.

Evidence-based medicine (EBM) is a conscientious, explicit, and judicious use of current best evidence in making decisions about the care of individual patients. The idea has now been expanded to encompass the relationship between patients and healthcare professionals in clinical practice, as well as the integration of patient preferences with the health care professionals experience and the best available research evidence in the decision-making process.

Generally, EBDM practice can be applied in five steps: step one; converting the current clinical scenario into an answerable question, step two; identifying the best available research evidence by performing a proper literature review across various databases, step three; critical appraisal of the evidence for its validity, impact, and applicability, step four; applying the results of the appraisal with the clinical experience and patient's values, and step five; evaluating the process and finding ways to improve it in the next time.

Applying evidence based descion making will improve diagnosis, clinical judgment, decision making and patient outcomes. Despite its utmost importance in developing countries for its cost-effectiveness and efficient use of healthcare resources, training need assessments has identified the need for the course for health professionals. Moreover, because of the importance of EBDM in building clinical practice on a scientific basis, there is a focus and a need on providing continuous training and capacity building programmes for health professionals. Evidentiary based skills and information accessible to clinicians by developing reliable clinical manual is also paramount.

The aim of this evidence based decision making course participant manual is to provide guidance, additional information, tools and resources to improve their knowledge, skill and attitude in providing evidence based clinical care. The training manual will also serve as a reference in all phases of the training. The manual is organized into five (5) chapters to address the most relevant themes in evidence based medicine and addresses gaps that were identified during the training needs assessment. Furthermore, each steps of evidence based medicine practice steps are include in detailed to improve participants' skills, knowledge and behaviors in integrating evidences into their decision making for individual patients.

Health care professional's failure to engage in EBDM has repercussions for the health of individuals and populations and it is strongly believed that the training will improve the evidence based clinical practice related gap. In addition, it improves both patient outcomes and trainees' practice-based learning and improvement. Different teaching modalities like interactive lectures, computer laboratory and clinical scenario based teachings and discussions will be used.

Rationale

EBDM is an approach that prompts professionals to use quantitative and qualitative data to develop informed plans. Evidence based decision making is essential for health professionals in order to provide the best care for their patients. It is important for health professionals to be aware of the latest research and best practices in order to make informed decisions that are tailored to the individual needs of patients.

Training on EBDM can help health professionals stay up-to-date on the latest research, understand the implications of different treatments, and make decisions that are in the best interest of their patients. EBDM training can also help to reduce medical errors, improve patient safety, and ensure the best outcomes for patients. With the right training, health professionals can ensure that their decisions are based on the most current, accurate, and reliable evidence.

Training health professionals in evidence-based decision-making can also help ensure that patients receive the best possible care. A training need assessment (TNA) conducted by EAA indicated that there is a gap among professionals in using Evidence based clinical decision making. Therefore, this training manual is prepared to encourage health professionals to apply evidence-based decision making in their clinical practice.

Target Audiences

This EBDM in-service training module is designed to equip all **health professionals** in the knowledge, skills and attitudes necessary to apply evidence based clinical decision-making practices.

Course syllabus

1. Module name: Evidence based Decision making

2. Module code:

3. Module duration: 5 days

4. Prerequisite: Health professionals with a minimum of BSc degree in their respective discipline

5. Module continuing education unit (CEU):

6. Module description: This four days training module/course is designed for all health professionals to equip them with all the essential knowledge, skills and attitudes on formulating relevant clinical questions, identifying hierarchy of evidence, conducting evidence retrieval, performing critical appraisal and incorporating up to date evidence in to clinical practice. The training will enable trainees make more informed decisions supported by the most current and up to date information which lead to better patient outcomes.

7. Module/course goal: at the end of this training, trainees will be able to make a sound clinical decision based on principles of evidence-based medicine.

8. Enabling/learning objectives

- Formulate clinically answerable questions
- Identify appropriate sources of evidence
- Critically appraise selected evidences to determine its validity, relevance, and applicability
- Implement EBDM principles in everyday clinical practice
- Evaluate the outcomes of interventions and clinical decisions

9. Chapters/sessions outline

Table 1: Chapter/session outline

Chapter	Title /topic	Durations	Sessions
1	Introduction for EBM and Formulating Answerable Clinical Questions	**5 Hrs**	• Introduction to Evidence Based Clinical Decision Making (EBCDM)(2 Hr) ○ Definition ○ Benefits of EBM ○ Standards and criterias for EBM • Formulating an Answerable Clinical Question (3Hrs) ○ Introduction ○ How to ask well formulated answerable questions? ○ The type of question being asked ○ The type of study that best answers the question
2	Evidence retrieval	**13 Hrs**	• Study Design and Strength of Evidence for Relevant Clinical Question (3 Hrs) • Sources of evidence (4 hrs) ○ The Haynes 5S knowledge acquisition pyramid ○ Study designs and Strength of Evidence • Effective Search Strategy (2 hrs) • Effective Database search (4 Hrs)
3	Appraising the evidence	**11 hour**	• Critical Appraisal (4Hrs) • Critical Appraisal Checklist (7)
4	Applying Evidence to Practice	**5 Hrs**	• Clinical Pathways and Decision analysis (3Hrs) ○ Clinical Pathways ○ Practice Guidelines ○ Decision Analysis • Assessing Recommendations and Facilitating Implementation of Evidence Based Practice (2Hrs) ○ Assessing Recommendations ○ Fascilitator and Barriers to the Implementation of Evidence into Practice
5	Assess Outcomes, Make Changes to Practice, and Quality Assurance	**4 Hrs**	Assess Outcomes and Make Changes to Practice (1 Hr) ○ Introduction ○ How to measure changes ○ Disseminating the results of an EBP intervention ○ How to assess your own EBP performance • Quality Assurance (1 Hr) ○ Quality indicators ○ Quality indicator assessment

10. Teaching learning methods and activities

- Interactive Lecture
- Group work
- Case studies/presentation
- Clinical scenario/group work/presentation
- Journal club critical appraisal/individual work/presentation
- Database searching presentation
- Role play and simulation (e.g. communicating evidence to patient)
- Hands on exercise (e.g. writing clinical questions)

11. Learning materials and resource (including references)

- Training manual
- Scientific journals and articles
- EBDM text books
- Access to research databases like Cochrane, HINARI or Pubmed database
- Prepared case studies/case scenarios
- slides displaying research study flow, baseline data, and results of the EBDM topic in discussion
- videos or webinars (application of EBDM on real world practice)
- Critical appraisal guidelines
- Clinical guidelines
- EBDM check list for implementation
- Writing boards
- Posters/pictures
- Laptops
- White board, marker
- LCD projector
- PPT

12. Reference

- *David L. Sackett, Sharon E.. Straus, Richardson WS. Evidence-based medicine: how to practice and teach EBM. Churchill Livingstone; 2000.*
- *Prasad K. Fundamentals of evidence based medicine. New York: Springer; 2013.*
- *Gøtzsche P. Rational diagnosis and treatment: Evidence-based clinical decision-making. John Wiley & Sons; 2008 Mar 11.*
- *Fleisher LA. Evidence-Based Practice of Anesthesiology E-Book. Elsevier Health Sciences; 2013 Jan 15.*
- *Hecht L, Buhse S, Meyer G. Effectiveness of training in evidence-based medicine skills for healthcare professionals: a systematic review. BMC medical education. 2016 Dec;16(1):1-6.*
- *Zwolsman SE, van Dijk N, Te Pas E, Wieringa-de Waard M. Barriers to the use of evidence-based medicine: knowledge and skills, attitude, and external factors. Perspectives on medical education. 2013 Feb;2(1):4-13.*
- *Shaneyfelt T, Baum KD, Bell D, Feldstein D, Houston TK, Kaatz S, Whelan C, Green M. Instruments for evaluating education in evidence-based practice: a systematic review. Jama. 2006 Sep 6;296(9):1116-27.*
- *Shafaghat T, Bastani P, Nasab MH, Bahrami MA, Montazer MR, Zarchi MK, Edirippulige S. A framework of evidence-based decision-making in health system management: a best-fit framework synthesis. Archives of Public Health. 2022 Dec;80 (1):1-20.*
- *Greenhalgh T. How to read a paper: The basics of evidence-based medicine. John Wiley & Sons; 2014 Apr 7.*

13. Learning Assessment methods

- Pre & posttest
- Participant feed back
- Quiz, exercises

14. Trainer and facilitators selection criteria

- Facilitators will be those who have developed this manual and those who undertaken the trainers course and have TOT on evidence based decision making

15. Trainees and participants selection criteria

- Health care professional providing clinical service in different units.

16. Training evaluation methods (satisfaction, learning, behavior and impact)

- Surveys
- Interviews
- Observation
- Pre and posttest

17. Venue - All CPD accredited institutions

18. Composition

19. Schedule

Table 2: Training Schedule

Time (LT)	Day 1	Day 2	Day 3	Day 4	Day 5
2:30-3:30	○ Welcome and introduction ○ Participant registration ○ Self-introduction and expectation ○ Pretest (MCQ)	**Warm up and agenda Cont.** **Evidence retrieval** ○ Study Design and Strength of Evidence for Relevant Clinical Question ○ Study design types ***Activities***	**Warm up and agenda Cont.** **Evidence retrieval** **Sources of evidence** ○ The knowledge acquisition pyramid ○ filtered and unfiltered sources of evidence ***Activities***	**Warm up and agenda Cont.** **Appraising the Evidence** ○ Critical Appraisal Checklist ***Activities***	**Warm up and agenda Cont.** **Applying Evidences to Practice** ○ Clinical Pathways and Decision analysis ***Activities***
3:30-4:30	**Introduction to Evidence Based Clinical Decision Making** • Definition • Benefits of EBM • Standards and criterias for EB ***Activities***		**Evidence retrieval** ○ Effective Database search ***Activities***	**Appraising the Evidence** ○ Critical Appraisal Checklist ***Activities***	**Applying Evidences to Practice** ○ Clinical Pathways and Decision analysis ***Activities***
4:45-6:30	**Introduction to Evidence Based Clinical Decision Making** ○ Standards and criterias for EBM	**Evidence retrieval** ○ Study Design and Strength of Evidence for Relevant Clinical Question	**Evidence retrieval** ○ Effective Database search	**Appraising the Evidence** ○ Critical Appraisal Checklist	**Applying Evidences to Practice** ○ Assessing Recommendatio

	Formulating an Answerable Clinical Question ○ Introduction ○ How to ask well formulated answerable questions? ***Activities***	○ Identifying types of study design for clinical question ***Activities***	***Activities***	***Activities***	ns and Facilitating Implementation of EBP ***Activities***
LUNCH	Lunch break				
8:00-9:30	**Formulating an Answerable Clinical Question** ○ The type of question being asked ○ The type of study that best answers the question ***Activities***	**Evidence retrieval Effective Search Strategy** ○ ***Activities***	**Appraising the Evidence** ○ Critical Appraisal ***Activities***	**Appraising the Evidence** ○ Critical Appraisal Checklist ***Activities***	Assess Outcomes, Make Changes to Practice, and Quality Assurance ***Activities***
9:00-9:45	**Formulating an Answerable Clinical Question** ○ The type of question being asked ○ The type of study that best answers the question ***Activities***	**Sources of evidence** ○ The knowledge acquisition pyramid ○ filtered and unfiltered sources of evidence ***Activities***	**Appraising the Evidence** ○ Critical Appraisal ***Activities***	**Appraising the Evidence** ○ Critical Appraisal Checklist ***Activities***	Post test
9:45-11:00			**Appraising the Evidence** ○ Critical Appraisal ***Activities***	**Appraising the Evidence** ○ Critical Appraisal Checklist***Activities***	**Feed back and closing**

Chapter One: Introduction for EBM and Formulating Answerable Clinical Questions

Duration: 4 hours

Chapter Description: This chapter is designed to give an overview about evidence based clinical decision-making process and also to enable trainees formulate an answerable clinical question as the first step of evidence-based medicine. It discusses about the benefit of EBM, standards and criteria for EBM, steps of EBM, levels of evidences and formulating good clinical questions. Making a well formulated answerable clinical question is a key step of evidence-based medicine as not well formulated questions will make finding answers in the literature difficult.

Primary Objective: The objective of this chapter is to provide background information about Evidence Based Clinical Decision Making and enable trainees to formulate relevant answerable clinical questions.

Specific Objectives: At the end of this chapter the trainees will be able to:

- Define evidence based clinical decision making (EBCDM)
- Explain the benefits of EBCDM
- Describe the 5 steps of EBCDM / EBM
- Categorize evidences in to different levels based on their strength of evidence
- Formulate an answerable relevant clinical question using the PICO (T) framework
- Differentiate the type of question being asked
- Identify the type of study which best answers the formulated question

Outline;

1.1 Introduction to Evidence Based Clinical Decision Making

1.2 Formulating an answerable clinical question

1.1: Introduction to Evidence Based Clinical Decision Making (EBCDM)

Duration: 1Hr

Session Objective: At the end of this session, trainees will be able to:

- Define evidence based clinical decision making (EBCDM)

- Explain the benefits of EBCDM
- Describe the 5 steps of EBCDM / EBM
- Categorize evidences in to different levels based on their strength of evidence

Activity1.1.1: Think-Pair-Share

1. What is Evidence based medicine?
2. What are the benefits of evidence based medicine

Duration – 20 min

Instruction

- T: (Think): "think" about what you know about the above 2 questions and write your answer - 5 minutes
- P: (Pair): pair with other participants or a small group (4 or 5 per group) and discuss your answers – 5 minutes
- S: (Share): share your thinking with the whole group. 10 minutes

1.1.1 Definition

Evidence-based medicine (EBM) can be defined as is an approach to medical practice intended to optimize decision-making by emphasizing the use of evidence from well-designed and well-conducted research. It is a thorough, explicit and judicious use of current best evidence in making decisions about the care of individual patient. The aim of EBM is to integrate the experience of the clinician, the values of the patient, and the best available scientific information to guide decision-making about clinical management.

Evidence-based medicine can also be defined as a medical approach that emphasizes the use of evidence from well-conducted scientific research to guide clinical decision-making. The goal of EBM is to provide practitioners with the most current and accurate information to inform their diagnosis, treatment, and management of patients. This approach involves critically appraising the available evidence, applying clinical expertise, and considering patients' values and preferences. EBM helps to ensure that healthcare decisions are based on the best available evidence, rather than relying on tradition, intuition, or personal experience. This approach is considered to be a key component of providing high-quality, safe, and effective healthcare.

1.1.2 Benefits of EBM

Evidence-based medicine (EBM) is a method of making medical decisions that involves using the best available evidence from scientific research to inform clinical practice. There are several key benefits to using EBM in healthcare.

EBM helps to ensure that patients receive the most effective and safe treatments. By relying on the best available scientific evidence, healthcare professionals can make informed decisions about which treatments are likely to be most effective for a given patient, based on the patient's individual characteristics and medical history. This can help to reduce the risk of adverse reactions and complications, and improve patient outcomes. Additionally, EBM helps to promote the efficient use of resources, by identifying treatments that are not only effective, but also cost-effective.

In addition, EBM helps to foster a culture of continuous learning and improvement within the healthcare system. By continuously evaluating and updating treatment protocols based on new scientific evidence, healthcare professionals can stay up-to-date on the latest advancements in medical knowledge and technology. This can help to improve the overall quality of care provided by the healthcare system, and ensure that patients receive the most advanced and effective treatments. Furthermore, using EBM can also lead to the identification of gaps in knowledge and areas where further research is needed, helping to drive forward scientific discoveries and advancements in medicine.

Also, EBM is a powerful tool that can be used to improve the quality, safety, and efficiency of healthcare, and is an important part of the modern healthcare system.

1.1.3 Standards and criterias for EBM

The standards and criteria for EBM are designed to ensure that decisions are made using the best available evidence, rather than relying solely on intuition, expert opinion, or tradition.

One of the key standards for EBM is the use of randomized controlled trials (RCTs) as the highest level of evidence. RCTs are designed to eliminate bias and confounding factors by randomly assigning participants to different treatment groups, making it easier to determine the cause-and-effect relationship between a treatment and its outcomes. Other types of evidence, such as observational studies, case reports, and expert opinion, are also considered but are generally considered to be of lower quality than RCTs.

EBM also involves assessing the risk of bias in the studies, looking at the statistical precision, generalizability of the results, and considering the patient's values and preferences.

The GRADE (Grading of Recommendations Assessment, Development and Evaluation) system is a widely-used tool for evaluating the quality of evidence and the strength of recommendations in clinical practice guidelines. It is a transparent, systematic and explicit approach that involves rating the quality of evidence according to certain criteria (such as study design, bias, consistency, and precision) and then using this information to determine the strength of recommendations. The strength of recommendations is classified as strong (in favor of an action) or weak (in favor of an action but with a lower level of certainty) based on the quality of the evidence supporting the recommendation and the balance of benefits and harms. The GRADE system is intended to help guideline developers and users make more informed decisions about the use of specific interventions in clinical practice.

By considering all these criteria, EBM ensures that the best available evidence is used to inform clinical practice, leading to better outcomes for patients.

By incorporating evidence from current scientific research, evidence-based medicine (EBM) seeks to effectively enhance the decision-making process in regards to healthcare. Science, the knowledge and experience of healthcare professionals, and the patient's personal values and beliefs are taken into consideration to identify the most beneficial medical choice for the individual patient.

There are 5 main steps for applying EBM to clinical practice:

- Defining a clinically relevant question
- Searching for the best evidence
- Critically appraising the evidence
- Applying the evidence
- Evaluating the performance of EBM

Because the evidence-based process informs future questions and practice, it is useful to imagine it as a continuous cycle. The following figure illustrates the 5 steps of EBM.

Figure 1: Steps of Evidence based Medicine

** Adopted from Evidence-Based Behavioral Practice, "Steps for Evidence-Based Behavioral Practice," http://www.ebbp.org/steps.html Accessed on Jan 28, 2023*

EBM starts with formulating a clinical question. The clinical question is an issue which the health-care provider addresses with the patient. After the clinical question is formulated, relevant scientific evidence is sought, which relates to the clinical question. Scientific evidence includes study outcomes and opinions. Not all data has the same strength. Recommendations from an expert are not as robust as the results of a well-conducted study, which is not as good as the result of a set of well-conducted studies. Thus in evidence-based medicine, the levels of evidence or data should be graded according to their relative strength. Stronger evidence should be given more weight when making clinical decisions.

The evidence is commonly stratified into six different levels:

- **Level IA:** evidence obtained from a meta-analysis of multiple, well-conducted, and well-designed randomized trials. Randomized trials provide some of the strongest clinical evidence, and if

these are repeated and the results combined in a meta-analysis, then the overall results are assumed to be even stronger.

- **Level IB:** evidence obtained from a single well-conducted and well-designed randomized controlled trial. The randomized controlled study, when well-designed and well-conducted, is a gold standard for clinical medicine.
- **Level IIA:** evidence from at least one well-designed and executed non-randomized controlled study. When randomization does not occur, there may be more bias introduced into the study.
- **Level IIB:** evidence from at least one well-designed case-control or cohort study. Not all clinical questions can be effectively or ethically studied with a randomized controlled study.
- **Level III:** evidence from at least one non-experimental study. Typically, level III evidence would include case series as well as not well-designed case-control or cohort studies.
- **Level IV:** expert opinions from respected authorities on the subject based on their clinical experience.

All clinical studies or scientific evidence can be classified into one of the above categories. The clinician must then use their professional, clinical experience to extrapolate the scientific evidence as it applies to the specific patient. Most clinical studies have specific inclusion and exclusion criteria, as well as the specific population studied. More often than not, the patient being treated by the clinician will have one or more substantial differences from the population in the study. The medical provider must then use their clinical judgment to determine how the variations between the patient and the study population are important or not and how they affect applying the study results to the specific patient.

For example, a specific patient may be a 70-year-old female with a history of hyperlipidemia and a new diagnosis of hypertension, looking at hypertension treatment options. The clinician may find a good randomized controlled trial looking at medications to control hypertension, but the study's inclusion criteria were a population of 18 to 65-year-olds.

- Should the clinician ignore the results as the specific patient does not meet the study demographics? Or
- Should the clinician ignore the age difference between the specific patient and study population?

This is where the clinical judgment helps bridge the gap between the relevant scientific evidence and the specific patient being treated.

Clinicians using evidence-based medicine must put all of the information in the context of the patient's values or preferences. The patient's values or preferences may conflict with some of the possible options. Even strong evidence supporting a specific treatment may not be compatible with the patient's preferences, and thus, the clinician may not recommend the treatment to the patient. Also, the treatment might not apply to the specific patient.

As an example, a patient may have a particular form of cancer. Level IA evidence may suggest life expectancy can double from 8 to 16 months with chemotherapy. The chemotherapy has significant side effects. The patient may find those side effects not acceptable and elect not to pursue chemotherapy secondary to the specific patient's preferences and values.

Once the clinical question is formulated, relevant scientific information is evaluated, and clinical judgment is used to apply the relevant scientific evidence to the specific patient and their values, the outcome must be evaluated. The final step is a re-evaluation of the patient and clinical outcome after application of the applied information. Did the intervention help? Were the outcomes as expected? What new information is obtained? How can this information be applied to future situations and patients?

Summary

Evidenced-based medicine starts with the clinical question and returns to the clinical question at the end to see to what effect the process worked. Without continuous re-evaluation, the medical provider will not be sure if their impact is positive or negative. Evidence-based medicine is a perpetual wheel of improvement rather than a one-time linear process.

1. 2: Formulating an Answerable Clinical Question

Duration: 3 hours

Objectives: At the end of this session, trainees are expected to:

- Formulate an answerable relevant clinical question
- Differentiate the type of question being asked
- Identify the type of study which best answers the formulated question

Activity 1.2.1: Think-Pair-Share
Duration – 20 min

1. Why is formulating a good question important in EBM
2. What question formulation formats do you know

Instruction

- T: (Think): "think" about what you know about the above 2 questions and write your answer - 5 minutes
- P: (Pair): pair with other participants or a small group (4 or 5 per group) and discuss your answers – 5 minutes
- S: (Share): share your thinking with the whole group. 10 minutes

1.2.1 Introduction

Evidence based Medicine starts with formulating a clinical question. Formulating a clinical question in EBM is an important step in the process of finding and evaluating relevant research. A well-formulated clinical question will help to guide your search for evidence and ensure that the information you find is relevant and useful to your specific situation. The **PICO** (Population, Intervention, Comparison, Outcome) framework is commonly used to help structure clinical questions.

The first step in formulating a clinical question is to identify the population of interest, such as a specific patient group or a disease state. Next, you should consider the intervention or treatment that you are interested in, such as a drug or surgical procedure. The comparison group can be a control group, a different treatment or no treatment. Finally, you should define the outcome that you are interested in, such as improvement in symptoms or reduction in mortality. By following this framework and being as specific as possible, you can develop a clear, focused clinical question that will help you find the most relevant and useful evidence to inform your clinical decision making.

1.2.2 How to ask well formulated answerable questions?

Questions are often only partly formulated, which makes finding answers in the literature a challenge. Breaking down the question into its component parts and restructuring it so that it is easier to find answers is an important first step in EBP.

Most clinical questions can be divided into four components, often abbreviated as **PICO**:

- **Patient, Population or Problem (P)**: What person or group of people are you interested in? What is the specific clinical problem that you have in mind?
- **Intervention or Indicator (I)**: What is the treatment strategy, exposure or test that you want to find out about in relation to the clinical problem? This might be:
    - An intervention: a procedure, such as a drug treatment, surgery or diet
    - An Indicator: exposure to an environmental hazard, a physical feature such as being overweight, or a factor that might influence a health outcome
    - An index test: a diagnostic test, such as a blood test or brain scan
- **Comparator or Control (C)**: an alternative control strategy, exposure or test.
- **Outcome (O)**: What are you or the patient most concerned about happening, or preventing happening?

A time frame is not applicable in some type of clinical questions, but it is sometimes useful to add the timeframe explicitly, giving us PICO (T):

- **Time (T):** What is the timeframe of the clinical question?

Activity1.2.2 : Case Scenario

You are presented with the following clinical problem: the parents of a severely autistic 6-year-old boy are unhappy that no interventions have significantly improved his lack of social communication. They were excited to learn from the parents of another autistic child that a new treatment - the use of intravenous secretin - has resulted in a dramatic benefit for many children.

Could this treatment help their child?

1. **What is the PICO (T) of this question?**
2. **What is the clinical question? (Formulate clinical question using PICO framework**

Duration – 45 mins

Instruction

- Make a group of 4/5 members
- Discuss on the scenario within groups for 15 minutes
- Share your answer to the whole group – 30 minutes

Once you have your clinical question in PICO (T) format, there are two additional facets that you should consider:

1. What type of question are you asking?
2. What type of study will best answer your question

1.2.3 The type of question being asked

There are different types of clinical question. The table below summarizes types of common clinical questions in EBM.

Table 3: Types of Common Clinical Questions in Evidence Based Medicine

Clinical Question Type	Description
Therapy / Intervention	• how to select interventions that will help a patient and that are worth the time and costs involved
Diagnosis	• How to determine whether a person has a condition or problem • How to select and interpret appropriate diagnostic tests
Etiology / Risk factor (cause of a disease)	• How to identify the cause of the disease • How to determine whether people with a given risk factor are more vulnerable to a condition or problem
Prognosis / prediction (probable course of a disease)	• How to predict a patient's clinical course into the future • How to anticipate potential complications
Frequency / Rate	• how to ascertain what percentage of the population has a condition or problem
Phenomena	• how to identify the outcomes most important to a patient or population

1. 2.4. The type of study that best answers the question

Different types of questions require different study designs. Once you have identified your question type, you will be better able to target the specific studies that best answer your clinical question.

In each case, a meta-analysis or systematic review of all relevant studies is preferable to an individual study.

The following table (Table 4) summarizes the best study designs for different type of study questions .

Table 4: Preferred study designs for different study questions

Question	Best study designs	Description
Therapy / Intervention	Randomized Controlled Trial (RCT)	Subjects are randomly allocated to treatment or control groups and outcomes assessed.

Etiology / Risk factor	RCT	As etiology questions are similar to intervention questions, the ideal study type is an RCT. However, it is usually not ethical or practical to conduct such a trial to assess harmful outcomes.
	Cohort Study	Outcomes are compared for groups with and without an exposure or risk factor: prospective study.
	Case-Control Study	Subjects with and without an outcome of interest are compared for previous exposure or risk factor: retrospective study.
Frequency and Rate	Cohort Study	As above
	Cross sectional study	Measurement of a condition in a representative, preferably random, population sample.
Diagnosis	Cross sectional study with random or consecutive sample	Preferably an independent, blind comparison with a gold standard test
Prognosis and Prediction	Cohort / Survival study	Long term follow-up of a representative cohort.

Activity 1.2.3 : Case Scenario

You are presented with the following clinical problem: the parents of a severely autistic 6-year-old boy are unhappy that no interventions have significantly improved his lack of social communication. They were excited to learn from the parents of another autistic child that a new treatment - the use of intravenous secretin - has resulted in a dramatic benefit for many children.

Could this treatment help their child?

1. **What is the question type?**
2. **What type of study will best answer an intervention question?**

Duration – 20 mins

Instruction

- Make a group of 4/5 members
- Discuss on the scenario within groups for 10 minutes
- Share your answer to the whole group – 10 minutes

Activity 1.2.4: Case Scenario

Your hospital is exploring diagnostic test options. The available tests for H1N1 flu influenza are Rapid antigen test and RT-PCR test. You want to use the test that accurately diagnose a recent H1N1 flu epidemic within a day in your area.

1. **What is the PICO(T) of this question?**
2. **What is the clinical question?**
3. **What is the question type?**
4. **What type of study will best answer a diagnosis question?**

Duration – 40 mins

Instruction

- Make a group of 4/5 members
- Discuss on the scenario within groups for 20 minutes
- Share your answer to the whole group – 20 minutes

Activity 1.2.5 : Case Scenario

A 25 years old pregnant mother visits the clinic with the complaint of severe headache, confusion, vision disturbance and chest pain. Her vital signs showed significantly increased BP. The available anti-hypertensive medications in the area are only Enalapril and Nifedipine. Which one of the 2 medications shall the physician order to the patient?

1. **What is the PICO of this question?**
2. **What is the clinical question?**
3. **What is the question type?**
4. **What type of study will best answer this question?**

Duration – 30 mins

Instruction

- Make a group of 4/5 members
- Discuss on the scenario within groups for 15 minutes
- Share your answer to the whole group – 15 minutes

Summary

Formulating a clear and specific clinical question is an important step in the evidence-based practice process. It helps to focus the search for relevant evidence and ensures that the information obtained is directly applicable to the patient's problem. The PICO framework (Patient/Population, Intervention, Comparison, Outcome) can be used to structure the question and ensure that all relevant components are included. It is also important to consider the type of question (e.g., diagnostic, prognostic, treatment, etiology, etc.) and the level of evidence needed (e.g. systematic review, randomized controlled trial, etc.) to guide the search for evidence. By taking the time to properly formulate a clinical question, healthcare providers can make informed decisions and provide the best possible care for their patients.

Chapter Two: Evidence retrieval

Duration: 12 hour

Chapter description: This chapter deals with equipping health professionals with the necessary knowledge and skill for how to search, retrieve and assess relevant sources to inform their clinical decision making. It covers the concepts of designing search, identifying study design and systematic database search.

Primary objective: at the end of this chapter, participant will be able to effectively and systematically search for evidence.

Enabling objectives: At the end of this chapter the trainees will be able to:

- Identify study designs for relevant clinical questions
- Formulate an effective search strategy for evidence
- Describe sources of evidence
- Perform efficient database search

Outline

2.1 study design and strength of evidence

2.2 effective search strategy

2.3 source of evidence

2.4 efficient database search

2.1: Study Design and Strength of Evidence for Relevant Clinical Question

Duration: 3 Hrs

Session objectives: by the end of this session trainees will be able to

- Types of study design
- Identify types of study design for clinical question

Activity 2.1.1: Think- Pair- Share

What kinds of study designs are available for clinical questions? What are their pros and cons?

Duration: 20 min

Instruction

- T: (Think): "think" about what you know about the above questions and write your answer - 5 minutes
- P: (Pair): pair with other participants or a small group (4 or 5 per group) and discuss your answers – 5 minutes
- S: (Share): share your thinking with the whole group-10 minutes

On average, clinicians have 5 to 8 questions about individual patients per daily shift. Approximately 2000 new articles are indexed in PubMed every day, and although few of them directly inform clinical practice, as many as 75 are randomized clinical trials and 11 are systematic reviews. Different types of questions require different study designs. Once you have identified your question type, you will be better able to target the specific studies that best answer your clinical question.

Activity 2.1.2: Case scenario

Identify the possible study design for the following clinical questions

- What is the incidence of nausea and vomiting after tympanoplasty surgery?
- What is the most effective treatment for increased intraocular pressure?
- What is the impact of obesity on pregnancy related complications?
- What is the association between caffeine consumption and the incidence of atrial fibrillation?

Duration – 30 mins

Instruction

- Make a group of 4/5 members
- Discuss on the scenario within groups for 15 minutes
- Share your answer to the whole group – 15 minutes

2.1.1 types of study design

Traditional study designs were classified as prospective and retrospective study design. Prospective studies begin at a time in the past and subjects are followed to the present time. retrospective studies begin at the present time and look back on the behavior or other characteristics of those subjects in the past.

Descriptive research

Studies that examine a number of cases or a cross-section of a population in order to seek for specific features are classified as descriptive studies since they serve as records of events. These are frequently

employed when a number of instances are recorded in which a novel treatment of a number of patients has encouraging outcomes, and the authors presenting the data want other doctors to be aware of the therapy. **Case reports** represent a single patient, whereas **case series** describe the experiences of a small number of patients with the same illness or therapy. **Cross-sectional studies** allow for simultaneous observation of all the intriguing characteristics of a patient population, including potential causes and effects.

Case studies and reports

The first description of a novel disease, clinical sign, symptom, treatment, or diagnostic test frequently comes from case reports or small numbers of patients. They may also be a description of a course of study, a procedure, a patient-care plan, or some other aspect of healthcare. A new sickness that is destined to become very important can be discovered by doctors thanks to some case reports. For instance, when the first cases were published in two case series in 1981, AIDS was first discovered.

Cross-sectional research

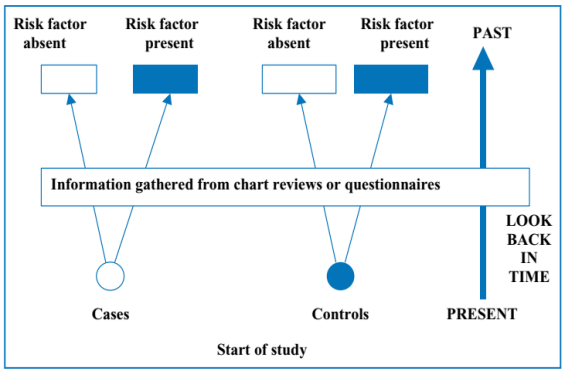
Cross-sectional studies are descriptive studies that examine a sample of a population to determine the prevalence of a specific disease and the prevalence of a specific risk factor. Cross-sectional studies keep track of activities and observations while describing illnesses, their origins, developments, consequences, or risk factors in a specific population at a particular point in time. Cross-sectional studies have the advantages of being generally rapid, simple, and inexpensive to conduct. Statistical databases or medical records are typically where you can find the data. They are helpful first exploratory investigations, particularly when screening or categorizing disease-related characteristics.

Longitudinal studies

Longitudinal study is a term describing either observations or interventions made over a given period of time. There are three basic longitudinal study designs: **case-control studies, cohort studies, and clinical trials**. These are analytic or inferential studies, meaning that they look for a statistical association between risk factors and outcomes.

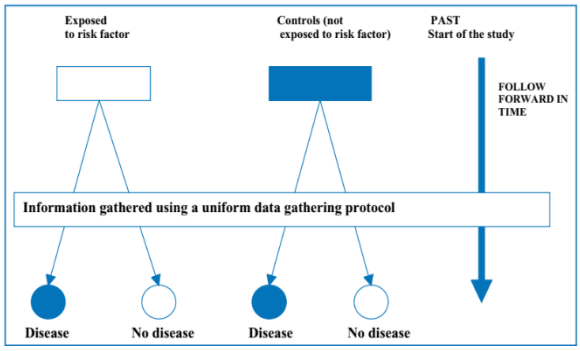
Case–control studies

These studies were previously called retrospective studies, but looking at data in hindsight is not the only attribute of a case–control study. The subjects are initially selected because they either have the outcome of interest – cases – or do not have the outcome of interest – controls. They are grouped at the start of the study by the presence or absence of the outcome, or in other words, are grouped as either cases or controls. This type of study is good to screen for potential risk factors of disease by reviewing elements that occurred in the past and comparing the outcomes.



Cohort research

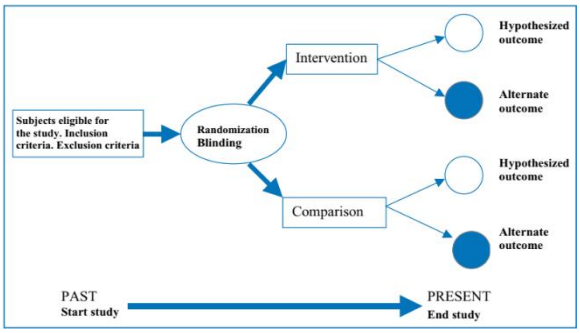
Because they are typically conducted from the past to the present, these investigations were traditionally referred to as prospective studies. The cohort is a collection of patients who have been chosen based on whether or not the risk factor is present. To ascertain which of them will experience the outcome or disease, they are monitored over time. For each group, it is possible to determine the incidence or risk, which is the likelihood of developing the outcome. The probability of developing the outcome is the incidence or risk, and can be calculated for each group. The degree of risk can then be compared between the two groups.



The cohort study can be one of the strongest research study designs. They can be powerful studies that can determine the incidence of disease and are able to show that the cause is associated with the effect more often than by chance alone. They can also show that the cause preceded the effect.

Clinical trials

A clinical trial is a cohort study in which the investigator intervenes by manipulating the presence or absence of the risk factor, usually a therapeutic maneuver. Clinical trials are human experiments, also called interventional studies. Traditional cohort and case–control studies are observational studies in which there is no intentional intervention.



Qualitative study design: answers a wide variety of questions related to human responses to actual or potential health problems. The purpose of qualitative research is to describe, explore and explain the health-related phenomena being studied.

Systematic review: A summary of the medical literature that uses explicit methods to perform a comprehensive literature search and critical appraisal of individual studies and that uses appropriate statistical techniques to combine these valid studies.

Meta-analysis: A systematic review that uses quantitative methods to synthesize and summarize the results.

Activity 2.1.3: Reflection activities

Duration – 30 mins

Instruction and question: Be in group of 2 and develop clinical questions for the following study designs

a) Case study
b) Cross sectional research
c) Case control
d) Cohort
e) Qualitative study design

2.1.2 Identify types of study design for clinical question

If you're unsure what the design of any given clinical question might be, the following table may prove helpful

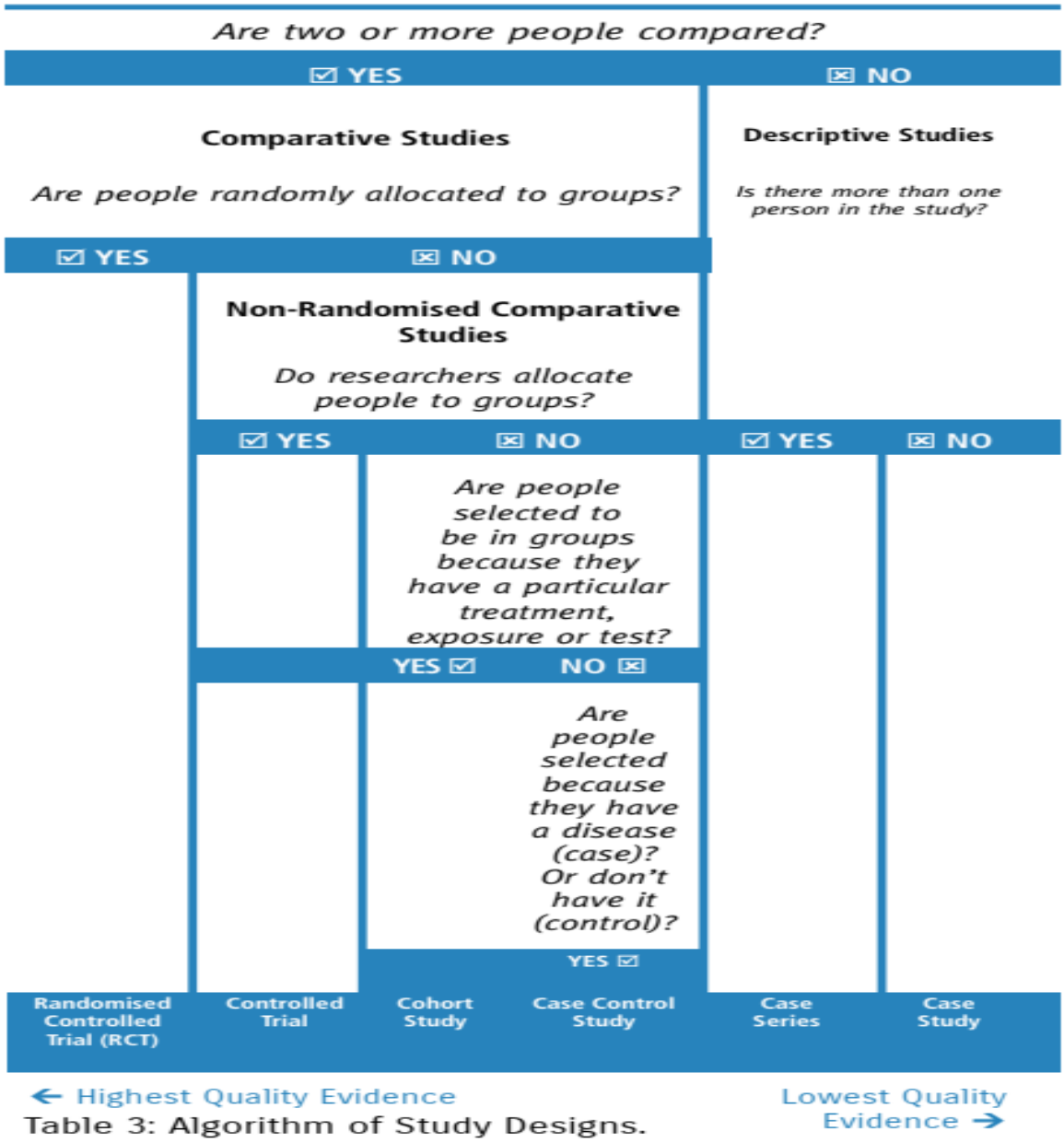


Figure 2: Algorism of Study Designs

Table 5: Types of Common Clinical Questions in Evidence Based Medicine

Clinical Question Type	Description
Therapy / Intervention	• how to select interventions that will help a patient and that are worth the time and costs involved
Diagnosis	• How to determine whether a person has a condition or problem • How to select and interpret appropriate diagnostic tests
Etiology / Risk factor (cause of a disease)	• How to identify the cause of the disease • How to determine whether people with a given risk factor are more vulnerable to a condition or problem
Prognosis / prediction (probable course of a disease)	• How to predict a patient's clinical course into the future • How to anticipate potential complications
Frequency / Rate	• how to ascertain what percentage of the population has a condition or problem
Phenomena	• how to identify the outcomes most important to a patient or population

Summary

After developing relevant clinical question, Identification of study design that is suitable for the proposed clinical question is important. This is primarily done by identifying the type of question asked in the developed clincal question (diagnosis, prognosis, therapy or harm). Study designs range from descriptive studies to meta-analysis and reviews. Identifying the strength and weakness of each study design for answering a given clinical question is paramount when searching for evidence. Comparative study quality increases as the number of people increase in between groups and as the assignment of those people become random.

2.2 Effective Search Strategy

Duration – 3 Hrs.

Session objectives: By the end of this session trainees will be able to

- Design search strategy
- Explain the methods of searching for evidence

Activity 2.2.1: Think-Pair-Share

- How do you search for reliable source of information?

- What key words can be used for searching for a randomized control trial of the effectiveness of lidocaine for alleviating post operative sore throat (therapy study)?
- What are the best ways to search for evidence on HIV/AIDS

Duration – 60 min

Instruction

- T: (Think): "think" about what you know about the above 2 questions and write your answer - 5 minutes
- P: (Pair): pair with other participants or a small group (4 or 5 per group) and discuss your answers – 5 minutes
- S: (Share): share your thinking with the whole group-10 minutes

After creating the clinical question, the search for evidence usually follows the following route

a. **Determine whether the question is a background or a foreground question.**

   Background questions, which usually start with "Who," "What," etc., demand a general understanding of a disorder. These are frequently questioned by novices or when patients are diagnosed with uncommon diseases. The greatest place to look when a query is about background information is frequently a textbook. The PICO framework is used to create the foreground questions, which are utilized to find particular information about how to manage patients with a problem. The majority of the time, evidence-based summaries or databases (such ACP Journal Club or Best Evidence) have the answers to the key questions (e.g. PubMed, EMBASE).

b. **Domain**

   Subsequently, choose the domain. Most health care articles can be classified into one of four domains: therapy, diagnosis, prognosis and harm/etiology.

c. **Search concepts**

   Before starting the search, map out the major concepts identified in the clinical question. This saves time, makes searching more efficient and can be done by using the attached worksheet. The key concepts are: “otitis media and antibiotics and child”.

   Using MeSH headings (Medical Subject Headings used to index articles according to their major topics) will help to narrow the search. In the MeSH Browser on PubMed, type in each of the concepts of your search and join them with AND.

d. Study design

As discussed in the above chapter, after determining the concepts, considering study design can be important. Particular questions may be best answered by a specific study design. Because the current question is a therapy question, systematic reviews, followed by well-conducted randomized controlled trials, are the highest form of evidence.

e. Sources

Following this, consider which sources to search. Depending on the question (and the study design), a variety of sources can be used. To be explained in the next chapter

Summary

Developing effective search strategy relies on determining whether the question is a background or a foreground question, identifying domains for the clinical question, using different search concepts like MeSH words for searching, identifying study designs and identifying the sources for those clinical questions.

2.3: Sources of evidence

Duration –5 hrs.

Session objectives

By the end of this session trainees will be able to

- Explain the knowledge acquisition pyramid
- Describe filtered and unfiltered sources of evidence

Activity 2.3.1: Group discussion

1. What is the knowledge acquisition pyramid and how can it help us understand EBM?
2. What are the different levels of knowledge in EBM and how can they be used to improve our practice?

Duration – 30 mins

Instruction

- Make a group of 5 members
- Discuss with groups for 15 minutes
- Share your answer to the whole group – 15 minutes

Before describing different sources of evidence, we need to discuss the hierarchy of evidence. This hierarchical system uses an adapted system of ranking different types of medical research based on their methodological qualities and research rigor. They arrange the evidence in descending order of strength.

2.3.1 The Haynes 5S knowledge acquisition pyramid

Most clinicians will want to do the most efficient searching at the point of care possible to aid the patient sitting in front of them, always turning to primary sources of evidence whenever a clinical question comes up is very inefficient at best and impossible for most busy practitioners. The busy clinician in need of rapid access to the most current literature requires quick access to high quality pre-appraised and summarized sources of evidence that can be accessed during a patient visit. This model which was developed by Dr. Steven Haynes describes the five levels of knowledge acquisition. This is a process of helping to find the best evidence at the point of care. The sources that are higher up on the appraisal pyramid (Fig. below) are the ones easiest to use and needing the least amount of critical appraisal by the user.

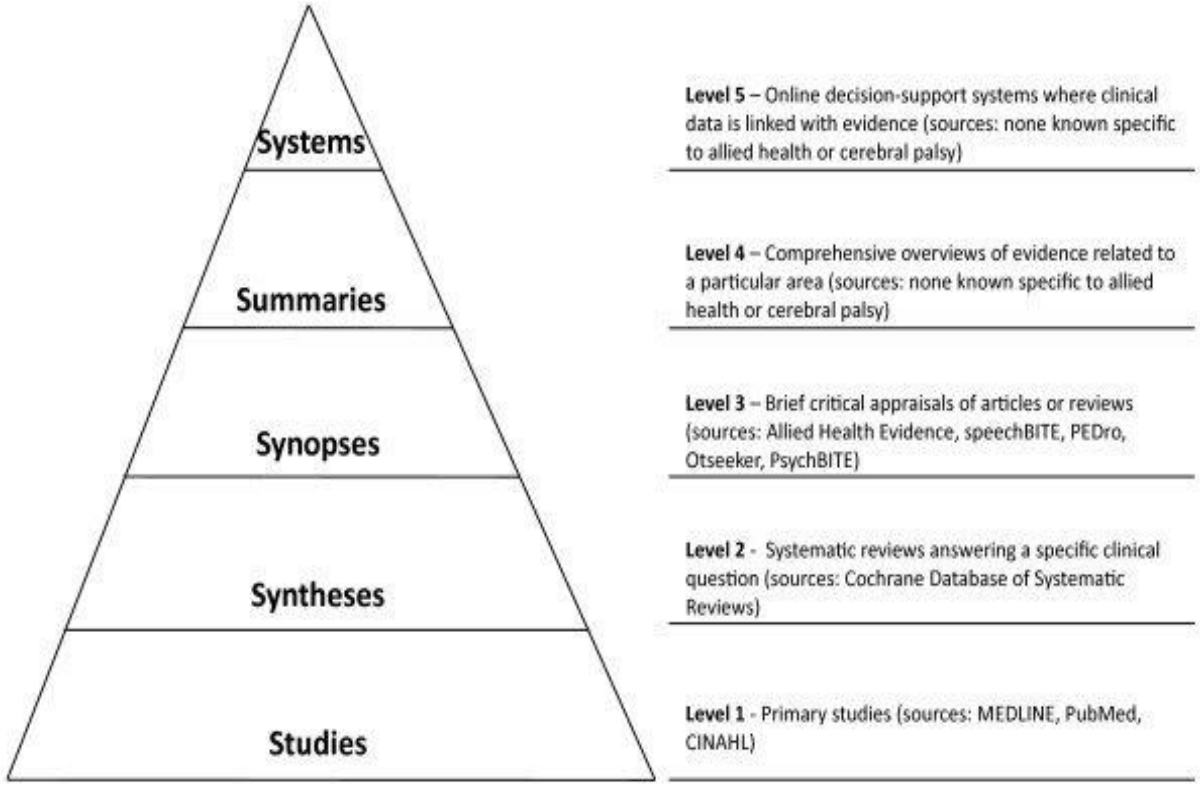
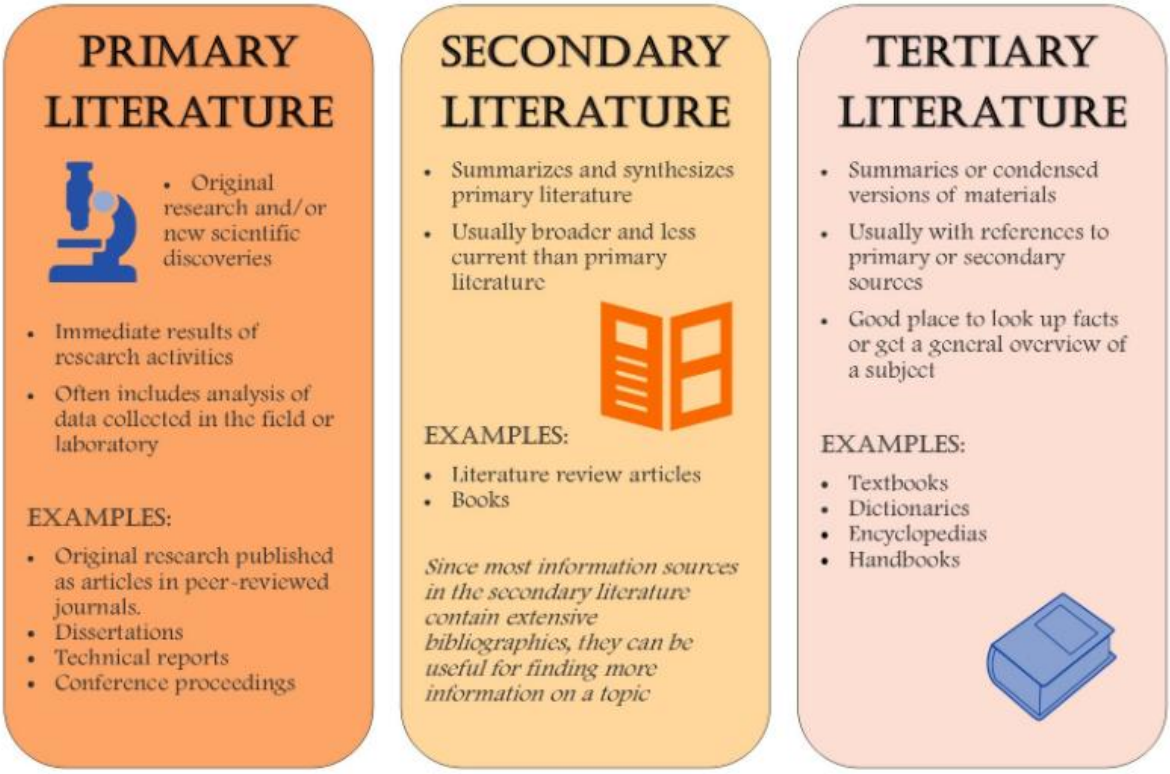


Figure 3: The Haynes knowledge acquisition pyramid

1. **Studies:** are primary or original studies, for example, those retrieved from MEDLINE/ PubMed. More than 20 million articles are indexed in MEDLINE. They need appraisal for clinical application.

2. **Syntheses** - are systematic reviews of the evidence relevant to a particular clinical question.
3. **Synopses** - provide brief critical appraisals of original articles and reviews. Examples of synopses appear in evidence-based journals such as ACP Journal Club (www.acpjc.org).
4. **Summaries -** provide comprehensive overviews of evidence related to a clinical problem (e.g., gout or asthma) by aggregating evidence from the lower levels of relevant synopses, syntheses and studies.
5. **Systems –** include such as electronic health records. At this level, clinical data are linked electronically with relevant evidence to support evidence-based decision-making. Computerized decision-support systems such as these are still rare, so usually we start at the second level from the top of the pyramid when searching for evidence.

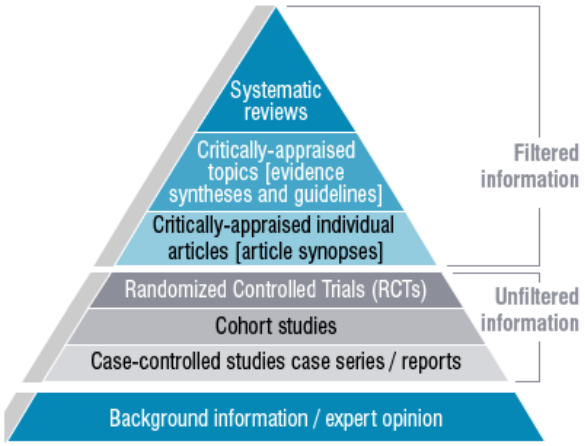


In evidence-based medicine, one way to organize the different types of evidence involved in evidence-based practice research is the levels of evidence pyramid described in the above table. primary literature is original research, such as peer-reviewed journal articles, that provides the most reliable evidence. Secondary literature is the interpretation, evaluation, and synthesis of primary literature, such as systematic reviews,

guidelines, and textbooks. Tertiary literature is the application of evidence to clinical practice, such as clinical practice guidelines, patient education materials, and decision support tools.

2.3.2 Filtered and unfiltered sources of evidence

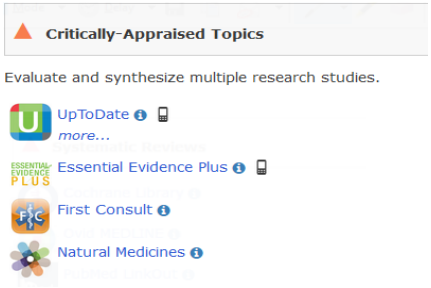
Filtered resources - are those that have been specifically selected and organized to provide clinical information that is relevant to a particular clinical question. These sources may include systematic reviews, meta-analyses, practice guidelines, and filtered bibliographic databases.



a) Systematic Reviews/Meta Analysis

TRIP (Turning Research Into Practice) Database Plus - Simultaneously searches evidence-based sources of systematic reviews, practice guidelines, and critically-appraised topics and articles -- including most of those listed above and many more. Also searches MEDLINE's Clinical Queries, medical image databases, e-textbooks, and patient information leaflets.

b) critically-appraised topics



EPC Reports - Using systematic review methods, EPCs review all relevant scientific literature on a wide spectrum of clinical topics to produce various types of evidence reports intended to inform health care decisions by patients, providers, policymakers, and other stakeholders.

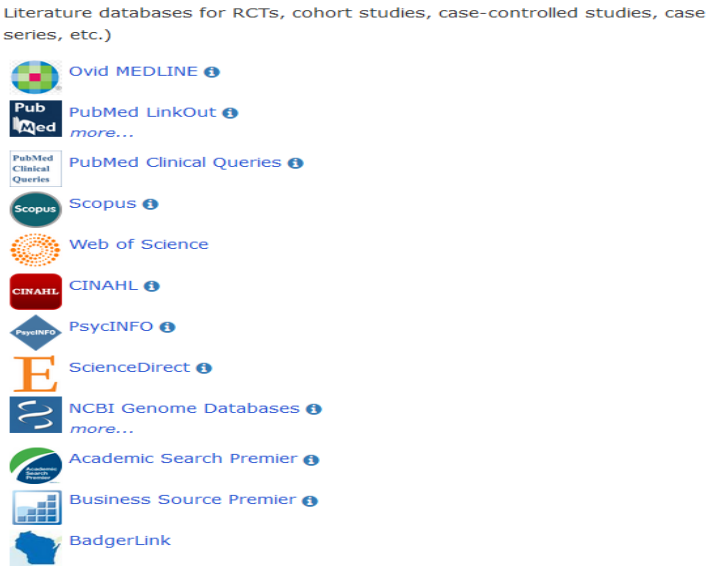
c) Critically-appraised individual articles

Such articles can be found in various databases, such as PubMed, EBSCOhost, and CINAHL. They can also be accessed through websites such as The Joanna Briggs Institute and The Centre for Evidence Based Medicine.

Unfiltered resources - are primary sources that describe original research. Randomized controlled trials, cohort studies, case-controlled studies, and case series/reports are considered unfiltered information.

a) **CINAHL® Plus** is the primary resource for the fields of nursing, allied health and occupational therapy. It includes: Evidence-Based Care Sheets, index/abstracts, full text articles, legal cases, clinical innovations, critical paths, drug records, research instruments and clinical trials. The database, CINAHL, is the core resource of this product.

b) **MEDLINE**, produced by the National Library of Medicine, is an important health science database covering, medicine, dentistry, and nursing. The resource uses Medical Subject Headings (MeSH) which allow for specific, focused searching. This version contains over 1,400 full text journals and is updated daily.



c) **PsycINFO** Provides access to journal articles and other materials in the field of psychology and psychological aspects of related fields. Coverage 1800s to present.

d) **PubMed** comprises more than 27 million citations for biomedical literature from MEDLINE, life science journals, and online books. Citations may include links to full-text content from PubMed Central and publisher web sites.

Background information, expert opinion

eMedicine - Medscape from WebMD offers specialists, primary care physicians, and other health professionals the Web's most robust and integrated medical information and educational tools.

Activity 2.3.2– pair up and search for evidence

- Does a low-carbohydrate diet result in better glycemic control compared to a conventional diet in pregnant women with gestational diabetes,
- Is physical therapy result better pain relief compared to medication alone for patients with chronic back pain?
- In patients with type 2 diabetes, does a diet and exercise program result in better glycemic control compared to a medication-only approach?
- What is the best treatment option for myalgia?

Duration: 60 minutes

Instruction: Pair up and search evidence for the above topics

Summary

The knowledge acquisition pyramid is useful for identifying strength of evidence. At the bottom of the pyramid are primary studies which are individual studies found on most websites like CINAHIL and PubMed. At the top of the pyramid are systems which are electronic health records which give the most wide and reliable evidence. Sources must be identified based on their information in to filtered and unfiltered sources. Filtered sources are those that have been specifically selected and organized to provide clinical information that is relevant to a particular clinical question. While unfiltered sources are those that provide access to all available literature on a given subject, including both relevant and irrelevant information.

2.4 Effective Database search

Duration: 2 Hrs

Session objectives: By the end of this session trainees will be able to

- Search for evidence efficiently

Activity 2.4.1: Think-pair-share

- What keywords should I use to find the best sources of evidence on current COVID treatment
- What search terms should I use to find evidence on malaria therapy?

Duration: 20

Instruction

- T: (Think): think about what you know about the above questions and write your answer - 5 minutes
- P: (Pair): pair with other participants or a small group (4 or 5 per group) and discuss your answers – 5 minutes
- S: (Share): share your thinking with the whole group-10 minutes

2.4.1 Effective database search

In chapter one we learned how to construct a well-built clinical question. Here we will be looking at how to convert a well-built clinical question into a strong search strategy for your literature search. The database search starts by Developing a list of search terms.

Developing a list of search terms

We start by Creating a list of search terms using elements from your PICO question. It may not be necessary to include all PICO elements in your search, but these key concepts are a good place to start. A PICO inquiry serves the straightforward function of identifying the search terms to be utilized to find the best evidence to address a pressing clinical question. In other words, the PICO question is the search strategy.

The search strategy leads to an unbiased and effective search. The unbiased and effective search leads to the evidence. The evidence answers the question and underpins the evidence-based recommendation, decision, or practice. Using a properly crafted PICO question while conducting a search enables for the discovery of a limited number of pertinent studies to address the query rather than unearthing hundreds of researches, the majority of which do not address the clinical question due to a badly prepared PICO question.

Example: What is the population, patient, or problem of interest? What is the main intervention or issue being considered? What outcome do you want to accomplish?

Consider synonyms, related terms, and acronyms for each key term. Using synonyms or related words can help you expand your search.

Example: If searching for traumatic brain injury, include "traumatic brain injury", "TBI", and "acquired brain injury".

In order to execute the search strategy, you have to fist determine which database(s) to search this will help in adjusting key words for example MeSH headings for PubMed, or CINAHIL, second Identifying the major elements of your question and third translating natural language terms to subject descriptors, MeSH Headings, or synonyms for example changing “Patients undergoing abdominal surgery” to “Abdominal Surgery”.

Set parameters for your search.

Combine keywords and phrases using terms such as "OR", “NOT” and "AND" (known as Boolean operators) to broaden or narrow your search results.

- “AND” retrieves results that include all the search terms. Use AND to limit your search and find evidence that must contain both words (e.g., "stroke AND aphasia"; "children AND hearing loss").
- “OR” retrieves results that include at least one of the search terms. Use OR to increase your search results and find evidence that contains either term (e.g., "dysphagia OR swallowing"; "teenagers OR adolescents").
- “NOT” excludes the retrieval of terms from your search.

Translating a PICO Question into Search Terms

Example Clinical Question—Does cognitive rehabilitation improve cognitive skills in adults with traumatic brain injury?

PICO	PICO element	Key words
P	Adult traumatic brain injury	Adult **AND** traumatic brain injury **OR** TBI OR brain injury
I	Cognitive rehabilitation	Cognitive rehabilitation **OR** cognitive training **OR** cognitive treatment
C	Not applicable	
O	Improved cognition	Cognition **OR** cognitive skills **OR** memory **OR** attention **OR** executive function **OR** problem solving

Each PICO Element (P, I, C, O) will be searched and saved individually using the correlating Search Strategy. In PubMed you can save citations temporarily using the Clipboard which provides a place to collect up to 500 items from one or more searches. After each search, you will clear the screen and start a new search before beginning your next search. After individually searching for all the PICO element now you have to combine the searches: To complete your search, you will combine the Population (those adult traumatic brain injury patients); the Intervention (Cognitive rehabilitation) with Outcome (Improved cognition). By using your database's Search History, you should be able to combine these searches into one search showing results from all three of your previous searches.

Activity

Think pair share

Convert the following clinical questions in to well-built search strategies

- In patients with osteoarthritis of the knee is hydrotherapy more effective than traditional physiotherapy in relieving pain?
- Does acupuncture help reduce pain and improve function in people with rheumatoid arthritis?
- Is cognitive behavioral therapy more effective than traditional pharmacological treatment for treating post-traumatic stress disorder?
- What is the most effective treatment for reducing symptoms of anxiety in children?
- Is there a difference between the effectiveness of antibiotics and antiviral medications in treating acute bronchitis?

Searching with MeSH – the medical subject Headings (MeSH) terms are a comprehensive controlled vocabulary used by the National Library of Medicine (NLM) to index articles in the field of biomedicine. You can mainly find MeSH terms at the NLM website https://meshb.nlm.nih.gov/

Use the MeSH database to find MeSH terms, including Subheadings, Publication Types, Supplementary Concepts and Pharmacological Actions, and then build a PubMed search.

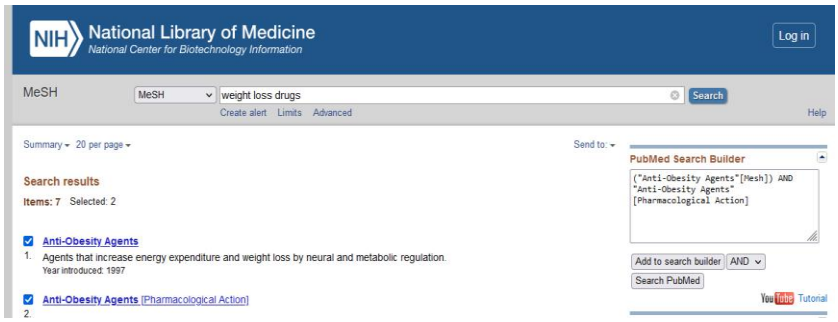
To build a PubMed search from MeSH:

1. Run a search in the MeSH database.
2. Select terms using the check boxes.
3. Click "Add to search builder" in the PubMed search builder portlet.

4. You may continue searching and including additional terms to the PubMed search builder using the "Add to search builder" and Boolean pull-down menu.
5. When you are finished, click "Search PubMed."

For example, < MeSH Database > right column – construct a search using the topic “weight loss drugs”

- Enter a topic, e.g. < weight loss drugs > maps to " Anti-Obesity Agents"[MeSH]; " Anti-Obesity Agents"[Pharmacological Action]
- Select MeSH and < Add to search builder >.



After adding all MeSH, use the Boolean operators like AND, OR as described above (and inserting parenthesis as needed) and finally search PubMed

Searching without mesh is often implemented in situations such as

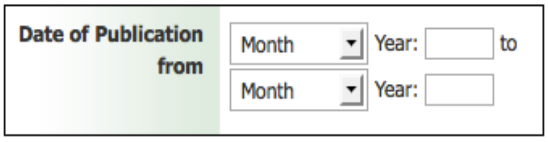
- New articles, not yet indexed
- Articles never indexed
- There is no MeSH for the topic (e.g., new concept; too specific, not biomedical)
- The indexer didn’t select the MeSH that you found most appropriate

In such a case a direct keyword-based searching will be implemented.

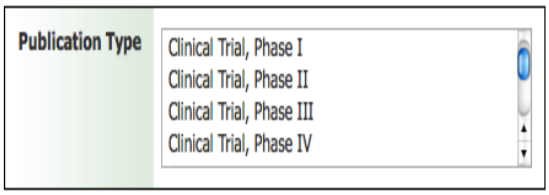
Refining results

Refine you results by adding limiters. Applying limiters to your search will allow you to focus your results to the most pertinent and relevant content ensuring that you aren’t wasting time wading through content that may not be useful. Search more effectively and efficiently by using the tools and limiters that are part of each database, apply limits and filters to narrow your search (e.g., date range, language).

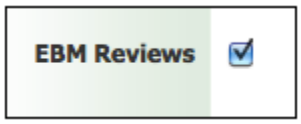
a) **Publication Date:** When doing research for EBM, you usually limit your search to articles from the last 3-5 years. A date limit may be helpful, particularly when a search retrieves too many results. Date limits may also be helpful if your question involves more recent technology or practice (e.g., ""digital hearing aids", "tele practice").



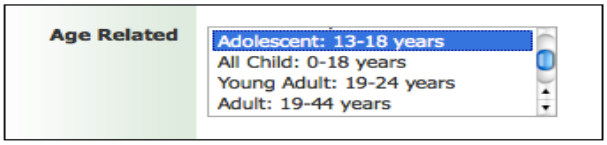
b) **Publication Type:** Limit your search to certain kinds of articles or research. Other options include comparative studies, systematic reviews, and randomized controlled trials.



c) **Reviews:** These present authors' findings after they have reviewed multiple previously-published research articles and can provide valuable information about trends in a field.



d) **Age Groups:** Limit by patient characteristics including age and sex.



e) **Species:** Species selections restrict your results to human or animal studies. In PubMed you can add species filters to the sidebar using the Additional Filters button.

f) **Language:** Language selections restrict your search to articles written in that language.

g) **Sex:** Sex restricts your search results to a specific sex for an animal or human study.

The final step to the perfect evidence search is to determine the level of evidence within each relevant article, based on the hierarchy of evidence lecture delivered to you in section 2.2.

Find a specific citation

To search for specific known author or title, paste the article title into the search box, or enter citation details such as the author, journal name and the year the article was published in the search box

- Enter the author's last name and initials without punctuation in the search box, and click Search.
- Searching for the journal by full entering journal title (e.g., molecular biology of the cell) or title abbreviation (e.g., mol biol cell) or ISSN number, a standardized international code

Summary

For Effective database search, researches should be categorized based on publication date, type and other criteria's such as age. The PICO search strategy uses the clinical question to develop an effective search strategy by combining with the Boolean operators. Other search strategies such as using MeSH headings

can also help simplify search results by categorizing results in to comprehensive index of medical terms. When the evidence is new and yet not indexed a direct search without the use of MeSH should be used.

Reference

1. *Guyatt G, Rennie D, Meade M. Users' guides to the medical literature. A manual for evidence-based clinical practice. Chicago: American Medical Association. 2002;705.*
2. *2.Leen B, Bell M, McQuillan P. Evidence-based practice: a practice manual.*
3. *Mayer D. Essential evidence-based medicine. Cambridge University Press; 2004 Jun 17.*
4. *Straus S, Haynes RB. Managing evidence-based knowledge: the need for reliable, relevant and readable resources. Cmaj. 2009 Apr 28;180(9):942-5.*
5. *Jensen KA. Steps To the Perfect Pico Search: Evidence-Based Nursing Practice. EBSCO Health. 7:1-9.*
6. *https://pubmed.ncbi.nlm.nih.gov/help/#combining-with-boolean-operators, accessed 2/09/2023*

Chapter Three: Appraising the Evidence

Duration: 11 hrs.

Chapter Description: To improve health care quality and expand evidence-based practice, the best available evidence should be used. As a result, carefully and methodically assessing the results of scientific research (evidence) is required. This session will provide a comprehensive overview of critical appraisal, the steps used in critical appraisal in a clinical setting, and a strong emphasis on the use of various tools to conduct thorough critical appraisal.

Primary Objective: After completing this chapter, the trainees will be able to critically appraise evidence and apply it in clinical settings.

Enabling objectives: At the end of this chapter the trainees will be able to:
▪ Outline the concepts of critical appraisal. ▪ Use methods for identifying valid evidence. ▪ Apply critical appraisal tools for therapy, prognosis, and diagnosis
Outline
3.1 Critical appraisal
3.2 Methods to Identify Valid, Relevance, Applicable Evidence.
3.3 Critical appraisal tools for therapy, prognosis, and diagnosis studies

3.1 Critical Appraisal

Duration: 4 Hrs

Session objective: At the end of this chapter the trainees will be able to:

- Outline the concepts of critical appraisal.
- Use methods for identifying valid evidence.

Activity 3.1.1: Think-Pair-Share

1. What is Critical appraisal? Discuss the concepts of Critical appraisal
2. Discuss the major components of critical appraisal

Duration – 30 min

Instruction

- T: (Think): "think" about what you know about the above 2 questions and write your answer – 5 minutes
- P: (Pair): pair with other participants or a small group (4 or 5 per group) and discuss your answers – 15 minutes
- S: (Share): share your thinking with the whole group. 10 minutes

Critical appraisal: is the process of carefully and systematically assessing the outcome of scientific research (evidence) to judge its trustworthiness, value and relevance in a particular context. Critical appraisal looks at the way a study is conducted and examines factors such as internal validity, generalizability, and relevance. Critical appraisal of the evidence involves three components.

Ask yourself:

1. What is the PICO(T) of the study and is it close enough to the PICO(T) of your clinical question?
2. How well was the study done? Is the quality of the study good enough to produce results that can be used to inform clinical decisions?
3. What are the results and are they applicable to your patients and your clinical setting?

1. What is the PICO of the study, and is it close enough to your PICO?

When determining whether a study will correspond exactly to your clinical question, it is critical to consider the PICO of the study in relation to the PICO of your clinical question. PICO stands for Population, Intervention, Comparison, and Outcome. A study is unlikely to provide an exact answer to your clinical question. You must decide whether it is close enough to answer your question: is the PICO of the study similar to the PICO of your clinical question? Is the study's population, for example, comparable to your patient or population? What outcomes are measured in the study, and do they correspond to the outcomes you are most concerned about achieving or avoiding? Once you've determined that the PICO of the study is close enough to the PICO of your clinical question, you can move on to the next question in the critical appraisal process.

2. How well - or how badly - was the study conducted?

The quality or internal validity of a study can be determined by asking yourself to what extent the research methods used minimized bias or other confounding factors. Bias can be defined as the systematic deviation of a study's results from the truth as a result of how it was conducted, analyzed, or reported. When "systematic error is introduced into sampling or testing by selecting or encouraging one outcome or answer over others," bias occurs. Bias can occur at any stage of a study, from study design to data collection and interpretation. Some examples of bias are provided in the table below (table:4.1).

Bias: *the systematic introduction of error into sampling or testing by selecting or encouraging one outcome or answer above others.*

Confounding: *the distortion of the true effect of a treatment, exposure or risk factor by other factors that vary from the study group to the control group. Randomization helps to eliminate confounding factors because the act of randomization distributes all confounders - both known and unknown - fairly*

Table 6: Types of Bias

Bias Type	**Description**
Selection or Sampling Bias	Systematic differences between those selected to participate in a study and those not selected.
Allocation Bias	Systematic differences in the allocation of participants to intervention and control groups.
Attrition Bias	Participants withdrawing from a study may differ systematically between intervention and control groups
Measurement Bias	Systematic differences in the measurement of an exposure or outcome between intervention and control groups.

Bias cannot be limited to a simple question: “Is bias present or not?” Instead, you must consider the degree to which bias was prevented by proper study design and implementation. There is no perfect study. As some degree of bias is nearly always present in a published study, you must consider to what extent bias may have influenced the results of a study. To determine how well bias and confounding factors have been avoided, each aspect of the study should be carefully scrutinized.

Ask yourself:

a. How were the subjects recruited?
b. How were the subjects allocated to groups?
c. How were the study groups maintained? Was there equal management and follow-up of subjects?
d. How were outcomes measured?

3. **What are the results and are they applicable to my patients and my clinical setting?**

   When you decide that the internal validity of a study is adequate and that bias and other confounding factors have been avoided, you need to closely examine the results of the study.

   Ask yourself:

   a. Are your patients similar enough to those in the study population that the results are applicable to your clinical setting?
   b. Did the intervention have a large enough effect on the clinical outcome(s) of interest that you would consider altering your practice and using the new intervention?
   c. What resources - human, financial, time - are needed to implement a change in clinical practice?

3.2: Critical Appraisal Checklist

Session Duration: 7 hrs

Session objective: at the end of this session, trainees are expected to utilize critical appraisal tools appropriately to critically appraise evidences.

Specific Objectives:

- Critical appraise a RCT and observational studies using the checklist
- Critically appraise a systematic review article using a checklist

3.2.1 Introduction

Critical appraisal checklists are designed to help assess the strengths and weaknesses of a study's design, methodology, and results. Checklists typically include a set of questions or criteria that must be answered or met in order for the study to be considered of high quality. This includes questions about the study's sample size, randomization process, control group, and data analysis methods, among others.

Using a critical appraisal checklist can help ensure that the evidence being used to inform decision making is of high quality and free from bias. This is especially important in healthcare, where the results of research

studies can have a significant impact on patient care and treatment decisions. By using a critical appraisal checklist, healthcare providers and researchers can make informed decisions about which studies to trust and which to be cautious of. Additionally, critical appraisal checklists can help identify areas for improvement in future studies, leading to higher quality research and better patient outcomes.

Critical appraisal checklists are prepared based on the study types. On the following pages sample critical appraisal guides for randomized controlled trials and systematic reviews will be discussed.

3.2.2 Critical Appraisal of Randomized Controlled Trials

Table 7: Critical Appraisal of Randomized Controlled Trials

Critical Appraisal of Randomized Controlled Trials		
Was the assignment of patients to study and control groups randomized?		
What's best?	**Where do I find the information?**	**In this study?**
Centralized computer Randomization is the gold standard and often used in multi-center trials. Smaller studies may use an independent person to monitor randomization.	The METHODS section should describe how patients were allocated to groups and whether or not randomization was concealed.	Yes ☐ No ☐ Can't tell ☐
Were study and control groups similar at the start of the trial?		
What's best?	**Where do I find the information?**	**In this study?**
If randomization is successful both groups should be similar. The trial should state whether differences are statistically significant: i.e., p-values.	The RESULTS section should include baseline characteristics comparing groups against a number of variables: age, risk factors, etc.	Yes ☐ No ☐ Can't tell ☐
Apart from the intervention under investigation, were both groups treated equally?		
What's best?	**Where do I find the information?**	**In this study?**

Apart from the intervention, both groups should be treated equally.	Look in the METHODS section for a follow up schedule and permitted additional treatments.	Yes ☐ No ☐ Can't tell ☐
Were all patients accounted for and analyzed in the groups to which they were originally allocated?		
What's best?	**Where do I find the information?**	**In this study?**
Losses to follow-up should be minimal (<20%). Patients should be analyzed in the groups to which they were randomized: i.e. intention to treat analysis.	The RESULTS section should state how many patients were randomized and how many were included in the analysis.	Yes ☐ No ☐ Can't tell ☐
Were measures objective or were participants and clinicians kept "blind" to which treatment was being received.		
What's best?	**Where do I find the information?**	**In this study?**
Ideally, the study should be double blinded: i.e. both patients and researchers do not know the treatment allocation. If the outcome is objective – e.g death – blinding is less important; if the outcome is subjective - e.g. symptoms - blinding is critical.	Look in the METHODS section to see if there is mention of masking of Treatments. The METHODS section should describe how the outcome was assessed and whether or not the researchers are aware of the patients' treatment.	Yes ☐ No ☐ Can't tell ☐
How large is the treatment effect?		
Results may be expressed in many ways, including: relative risk, absolute risk reduction, relative risk reduction and number needed to treat.		How large is the treatment effect?
How precise is the estimate of treatment effect?		
We can gauge how close the estimate of treatment effect is to the true value by looking at confidence intervals		What are the confidence intervals?
Are the results applicable in my clinical setting?		
	Are my patients similar enough to those in the study?	

	Is the treatment feasible/ affordable? Are the potential benefits worth the potential risks to the patient and/or costs involved in implementing a change in practice?
Is the trial relevant to my clinical question?	**Yes No Can't Tell**
Does the trial address a clearly focused question?	**Yes No Can't Tell**

Adapted from the Centre for Evidence-Based Medicine "RCT Critical Appraisal sheet" at http://www.cebm.net/?o=1040

3.2.3 Critical Appraisal of systematic reviews

Table 8: Critical Appraisal of Systematic Reviews

Critical Appraisal of Systematic Reviews		
What question - PICO (T) - did the systematic review address?		
What's best?	**Where do I find the information?**	**In this study?**
The question should be clearly stated. The exposure - eg a therapy or diagnostic test - and outcomes of interest are often expressed as a simple relationship.	The title, abstract or last paragraph of the introduction should clearly state the question.	Yes ☐ No ☐ Can't tell ☐
Is it safe to say that important, relevant studies were not missed?		
What's best?	**Where do I find the information?**	**In this study?**
A comprehensive search strategy should include subject searching of all relevant databases, manual searching of reference lists and contact with experts. Searches should not be limited to English language only. A combination of	The METHODS section should describe the search strategy. The RESULTS section should indicate the number of studies reviewed and excluded – with reasons for exclusion.	Yes ☐ No ☐ Can't tell ☐

subject headings and keywords should be used.		
Were the criteria used to select articles for inclusion appropriate?		
What's best?	**Where do I find the information?**	**In this study?**
The inclusion or exclusion of studies should be clearly predefined. Eligibility criteria should be formulated on the basis of the patients, interventions and outcomes of interest. In many cases, study design will also be a key component.	The METHODS section should describe in detail inclusion and exclusion criteria.	Yes ☐ No ☐ Can't tell ☐
Were included studies sufficiently valid?		
What's best?	**Where do I find the information?**	**In this study?**
The article should describe how the quality of each study was assessed using predetermined criteria appropriate to the type of clinical question: eg randomization, blinding and completeness of follow-up.	The METHODS section should describe the assessment of quality and criteria used. The RESULTS section should provide information on the quality of the individual studies.	Yes ☐ No ☐ Can't tell ☐
Were the results similar from study to study?		
What's best?	**Where do I find the information?**	**In this study?**
Ideally, the results of included studies should be similar or homogenous. If heterogeneity is present, statistical significance should be estimated and possible reasons explored.	The RESULTS section should state whether or not the results are heterogeneous. To identify heterogeneity, you may visually assess the forest plot or perform a statistical test: the chi-squared test.	Yes ☐ No ☐ Can't tell ☐
What are the results?		

A systematic review provides a summary of data from a number of individual studies. If individual studies are similar, a statistical method called meta-analysis is used to combine results. The meta-analysis gives weighted values to each study according to its size. The results of individual studies should be expressed in a common way – eg relative risk, odds ratio or mean difference between groups – and are normally displayed in a figure called a forest plot.			
Is the review relevant to my clinical question?	**Yes**	**No**	**Can't Tell**
Does the review address a clearly focused question?	**Yes**	**No**	**Can't Tell**

Adapted from the Centre for Evidence-Based Medicine "Systematic Review Critical A appraisal Sheet" at http://www.cebm.net/index.aspx?o=1567

NB: *Additional Critical appraisal tools for different study designs are found at Centre for Evidence-Based Medicine: http://www.cebm.net*

Activity 3.2.1: Critical Appraisal

Duration: 5 hours
Instruction:

- Divide the trainees into groups
- Read and critically appraise the given article 30 minutes
- Present your finding for the whole group 30 minutes per group

NB: The articles and the checklists of all study types are annexed at the end of this document

Reference

1. Guyatt G, Rennie D, Meade M. Users' guides to the medical literature. A manual for evidence-based clinical practice. Chicago: American Medical Association. 2002;705.

2.Leen B, Bell M, McQuillan P. Evidence-based practice: a practice manual.

3. Mayer D. Essential evidence-based medicine. Cambridge University Press; 2004 Jun 17.

4. Straus S, Haynes RB. Managing evidence-based knowledge: the need for reliable, relevant and readable resources. Cmaj. 2009 Apr 28;180(9):942-5.

Chapter Four: Applying Evidences to Practice

Duration: 5 Hrs

Chapter description: This chapter deals with applying evidences to the daily clinical practice by using principles of clinical decision analysis (decision tree), clinical pathways, and use of current best evidence to offer advice or treatment. The module is delivered in the form of interactive lecture, discussion, computer lab exercises, and case scenario-based activities. Trainees' performance will be assessed by self reflection, review questions and structured posttest.

Primary objective: At the end of this module participants will be able to recognize principles of evidence based medicine application to make clinical decisions that are in the best interest of the patients taking into account the potential risks and benefits of any intervention.

Enabling objectives: At the end of this session participants will be able to:

- Define clinical decision analysis
- Discuss concepts of clinical decision analysis and pathway for effective application of evidence into practice
- Identify several models used to guide healthcare professionals in the successful implementation of evidence into practice.
- Describe facilitators and barriers for effective implementation of evidence based practice
- Provide evidence based care for individual patient, taking into account their values, preferences, and clinical context

Outline

4.1 Clinical pathway

4.2 Clinical descion analysis

4.3 Assessing recommendations

4.4 Facilitators and barriers to the implementation of evidence based practice

4.1 Clinical Pathways and Decision analysis

Session Duration: 3Hrs

Session description: Clinical pathways (CPWs) are a quality strategy tools used to guide evidence-based healthcare and that aims to maximizing patient safety and clinical efficiency. The clinical decision analysis (CDA) is also used to overcome complexity and uncertainty in medical problems. It allows decision-makers to apply principles of evidence-based medicine to reach on objective clinical decisions when faced with complex clinical situations

Specific Objectives: at the end of this session trainees are expected to

- Describe the clinical pathways and their role in maximizing patient safety and clinical efficiency
- Discuss concepts of clinical Decision analysis
- Perform decision analysis using decision tree approach

Activity 4.1.1: Think-Pair-Share

- What is clinical descion analysis
- What are the importance of clinical descion analysis and clinical pathways

Duration – 20min

Instruction

- T: (Think) about what you know or have learned about the above question- 5minutes
- P: (Pair) with another participant or a small group and discuss the points - 5minutes
- S: (Share) your thinking with your partner and facilitators expand the "share" into a whole-class discussion- 10minutes

4.1.1 Clinical Pathways

Clinical pathways (CPWs) are a quality strategy tools used to guide evidence-based healthcare that aims to maximizing patient safety and clinical efficiency. It should be developed based on the best available evidence, such as systematic review or clinical practice guidelines that they have the potential to streamline clinical practice for a specific group of patients undergoing a particular procedure.

CPWs can also be described as structured multidisciplinary care plans which detail essential steps in the care of patients with a specific clinical problem.

Characteristics of CPWs:

1. It is used to translate guidelines or evidence into local structures.
2. It details the steps in a course of treatment or care in a plan, pathway, algorithm, guideline or protocol
3. It aims to standardize care for a specific clinical problem, procedure or episode of healthcare in a specific population.

CPWs could be implemented in any area of health care and mainly focus on processes in relation to electiveness, patient-safety and/or patient-centeredness. They are also strongly linked to recommendations from clinical guidelines

Why should clinical pathways contribute to healthcare quality?

- Aligns clinical practice with guideline recommendations in order to provide high-quality care within an institution
- Serve as useful tools to reduce variations in clinical practice, thereby maximizing patient outcomes and clinical efficiency.
- To promote safe, evidence-based care by providing locally oriented recommendations for the management of a specific condition, disease or reason to demand healthcare.
- Contribute to the reduction of complications and treatment errors
- Structure the flow of services for a group of patients with a particular diagnosis or undergoing a particular procedure and they guide the patient through the treatment process.
- Supports the translation of clinical guideline recommendations or evidence available in other forms into local protocols and clinical practice.
- Whilst clinical guidelines provide generic recommendations, CPWs institutionalize best practices to bring evidence to the bedside for all the health professionals involved.

It is important to take the patient's particular needs and preferences into account when applying evidence to practice. The application of evidence-based practice ought to be personalized for each patient, taking into consideration their values, preferences, and clinical setting. This helps to ensure that the care provided is appropriate and effective for the individual patient. When putting evidence into practice, it is crucial to take into account both the possible dangers and advantages of any intervention.

Practice Guidelines

Practice guidelines are statements that include recommendations intended to optimize patient care. They are, ideally, informed by a systematic review of evidence and an assessment of the benefits and harms of alternative care options.

To make a recommendation, guideline panelists must define clinical questions, select the relevant outcome variables, retrieve and synthesize all of the relevant evidence, rate the confidence in the effect estimates, and, relying on a systematic approach but ultimately also on consensus, move from evidence to recommendations. To inform their audience, guideline panels should provide only their recommendations but also the key information on which their recommendations are based.

4.1.2 Decision Analysis

Decision analysis is a formal method that integrates the evidence regarding the beneficial and harmful effects of treatment options with the values or preferences associated with those effects. Clinical decision analyses are built as structured approaches (decision trees).

The clinical decision analysis (CDA) is a tool usually used to overcome complexity and uncertainty in medical problems. It allows decision-makers to apply principles of evidence-based medicine to reach on objective clinical decisions when faced with complex clinical situations. Depending on the methodology used, the CDA stages can be summarized as:

1. Designing a decision tree showing all instances that can occur in a particular situation
2. Securing the likelihood and outcome utility values for each instance by conducting a rigorous literature search
3. Calculating the probabilities of cumulative expectation
4. Performing a sensitivity analysis and assessing the variables needed for clinical decision-making

An evidence-based decision tree is a tool used to help make decisions based on the best available evidence. It is a visual representation of the decision-making process, which can be used to identify the most appropriate course of action. It is composed of nodes (which represent decisions or options), branches

(represent the outcomes of the decisions). The decision analysis also addresses the extent to which each of the outcome events is desirable or undesirable (utility). The combination of the probabilities and utilities allows the decision analyst to determine the relative value of each management option (Figure).

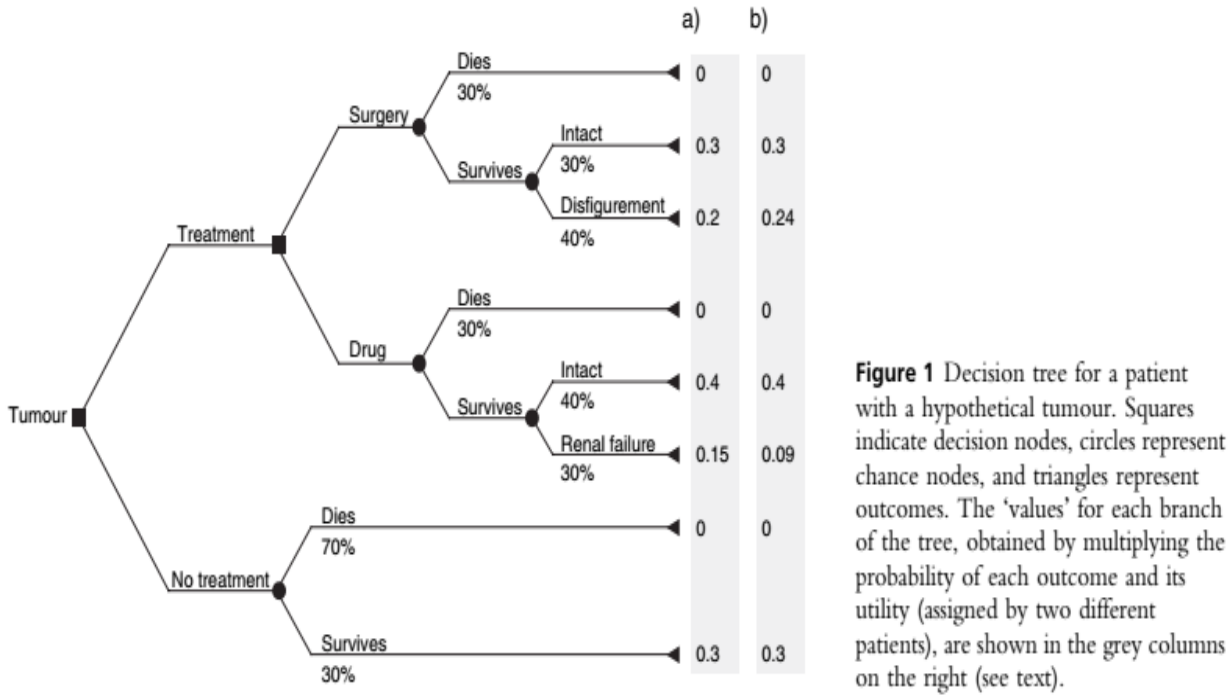


Figure 4: Sample Decision tree for a patient with hypothetical tumour.

Adopted from: Busse R, Klazinga N, Panteli D, Quentin W, World Health Organization. Improving healthcare quality in Europe: Characteristics, effectiveness and implementation of different strategies. World Health Organization. Regional Office for Europe; 2019

Summary

Clinical pathways (CPWs) are a quality strategy tools used to guide evidence-based healthcare that aims to maximizing patient safety and clinical efficiency. It should be developed based on the best available evidence, such as systematic review or clinical practice guidelines that they have the potential to streamline clinical practice for a specific group of patients undergoing a particular procedure. Furthermore, clinical decision analysis (CDA) is a tool usually used to overcome complexity and uncertainty in medical problems. It allows decision-makers to apply principles of evidence-based medicine to reach on objective clinical decisions when faced with complex clinical situations. Understanding the principles of clinical pathways (CPWs) and applying

clinical decision analysis (CDA) using the descion tree approach is very crucial for effective application or implementation of evidences to practice

4. 2 Assessing Recommendations and Facilitating Implementation of Evidence Based Practice

Session Duration: 3 Hrs

Session description: Assessing recommendation is an important part of the decision-making process. It involves evaluating the quality of the recommendation and determining whether it is suitable for the situation. Furthermore identifying the barriers and facilitators of evidence based practice implementation is very important for effective application of the identified practices.

Specific Objectives: at the end of this session trainees are expected to

- Describe the importance of assessing recommendation to evidence based clinical decision making
- Identify the barriers and facilitators of evidence based practice implementation
- Assess recommendations of different guidelines using criterias of recommendation assessment

Activity 4.2.1: Think-Pair-Share

What are the barriers and facilitators of evidence based practice implementation in your institutions?

Duration – 25min

Instruction

- T: (Think) about what you know or have learned about the above question- 7minutes
- P: (Pair) with another participant or a small group and discuss the points - 8minutes
- S: (Share) your thinking with your partner and facilitators expand the "share" into a whole-class discussion- 10minutes

4.2.1 Assessing Recommendations

Users' Guides for assessing Treatment recommendations

- Is the clinical question clear and comprehensive?
    - Is the recommended intervention clear and actionable?
    - Is the alternative clear?
    - Were all of the relevant outcomes important to patients explicitly considered?

- Was the recommendation based on the best current evidence?
- Are values and preferences associated with the outcomes appropriately specified?
- Do the authors indicate the strength of their recommendations?
- Is the evidence supporting the recommendation easily understood?
    - For strong recommendations, is the strength appropriate? For weak recommendations, does the information provided facilitate shared decision making?
- Was the influence of conflicts of interests minimized?

I. Is the clinical question clear and comprehensive?

The most practical recommendations for patient treatment from guidelines and decision analyses will follow a standard framework that specifically outlines the suggested actions, the alternatives with which they are compared, to whom and under what circumstances they apply.

Is the recommended intervention clear and actionable?

Recommendations are sometimes too vague to be helpful.

Example of vague recommendation: For both outpatients and inpatients with diabetic foot infection, clinicians should attempt to provide a well-coordinated approach by those with expertise in a variety of specialties, preferably by a multidisciplinary diabetic foot care team.

In this recommendation, the level of obligation (attempt) is not clear and the composition of the team is better described

Example of good recommendation: We recommend that a multidisciplinary foot care team manage the care of patients with diabetic foot problems who require inpatient care. Te multidisciplinary foot care team should include a diabetologist, a surgeon with the relevant expertise, a diabetes nurse specialist, a podiatrist and a tissue viability nurse.

Is the alternative clear?

When recommendations of guideline are developed by panelists, they usually choose a specific course of action over others. If the alternative is not clearly stated, the significance of the recommendation will remain obscure.

Were all of the relevant outcomes Important to patients explicitly Considered?

The balance between the harms and benefits of interventions will depend on what outcomes are considered. Clinicians should judge whether the guideline panel or the decision analysts included all patient-important outcomes which include mortality, morbidity (eg, major bleeding, acute exacerbation of a chronic disease, hospital admission), and patient-reported outcomes (eg, quality of life, functional status).

Were the recommendations based on the current best evidence?

Decision analysts should use recent or updated systematic reviews, ideally those that include meta-analysis, to inform their estimates of the benefits and harms of the intervention as well as their assessment of the associated confidence in effect estimates. In the absence of such meta-analytic systematic reviews, guideline panelists may conduct their own reviews or provide less systematic evidence summaries. Clinicians should look for a description of the process used to identify and summarize the relevant evidence and should judge to what extent this process is credible. Furthermore, recommendations that do not use the best current evidence may promote suboptimal or even harmful care.

II. Are values and preferences appropriately specified for each outcome?

Assessing treatment effects on outcomes is largely a question of measurement and a matter of science. Assigning preferences to outcomes is a matter of values.

Example: *In the outcomes associated with routine mammographic screening in women aged 40 to 49 years, there is a very small and questionable reduction of breast cancer mortality and a relatively high probability of a false-positive result (which leads to unnecessary follow-up testing and biopsy).*

The value attached to each of these 2 outcomes (reduction of breast cancer mortality and false-positive result) should be considered when trading them to develop a recommendation. A panel that assigns a higher value to the very small reduction in cancer mortality would support the screening, whereas a panel that assigns a higher value to avoiding unnecessary procedures would not. In addition, clinicians should look for explicit statements regarding the preferences and values used to inform the recommendation. Under ideal circumstances, recommendations should be based on a systematic review of relevant studies exploring patients' values and preferences. In the absence of a body of empirical evidence about patients' values and preferences, the experience of clinicians who regularly engage in shared decision making may be considered.

III. Do the authors indicate the strength of their recommendations?

Trustworthy recommendations should specify the strength of the recommendations and also a rating of the confidence in effect estimates that support the recommendations (quality of evidence). Sensitivity analyses are used to explore the strength of the conclusions that arise from a decision analysis. The 3 most commonly used approaches are GRADE (Grading of Recommendations Assessment, Development and Evaluation) and those used by the American Heart Association (AHA) and the US Preventive Services Task Force (USPSTF).

The 3 systems feature a rating for confidence in effect estimates (I e, quality of evidence). Confidence in the effect estimates represents the extent to which the estimates are sufficiently credible to support a particular recommendation. The GRADE approach specifies 4 levels of confidence: high, moderate, low, and very low. Whereas the AHA and USPSTF systems specify 3 levels of confidence: A, B, and C in the AHA approach and high, moderate, and low in the USPSTF approach.

The 3 systems share another critical feature: they differentiate between recommendations that should be applied (or avoided) in all, or almost all, patients (ie, strong recommendations) from those that require individualization to the patient's values, preferences, and circumstances (I e, weak recommendations).

Sensitivity analysis and grades of recommendation in clinical studies are important tools for assessing the quality of evidence and the strength of recommendations. Sensitivity analysis is a statistical technique used to assess the impact of different factors on the results of a study. It is used to identify which factors are most important in determining the outcome of a study and to determine the degree to which the results of a study are affected by changes in those factors. Grades of recommendation are used to assess the strength of a recommendation based on the quality of evidence.

IV. Is the evidence supporting the recommendations easily understood?

The message to the clinician from strong recommendations is "just do it." Recommendations that are inappropriately graded as strong may therefore have substantial undesirable consequences. High confidence in the effect estimates will support a strong recommendation if the desirable consequences considerably outweigh the undesirable consequences. When there is substantial uncertainty regarding the effects of the intervention (low confidence in the effect estimates), clinicians should generally expect weak

recommendations. Sometimes, guideline panels can appropriately offer strong recommendations despite low or very low confidence in effect estimates. Recommendations, in particular, weak recommendations should explicitly provide the key underlying information necessary to act on the recommendation.

V. Was the Influence of Conflicts of Interests minimized?

The judgments involved in the interpretation of the evidence and the decision on the final recommendation may be vulnerable to conflicts of interest. Clinicians can check the conflict of interest statements of the guideline panelists or decision analysts, usually found at the beginning or end of a publication or in a supplementary file

4.2.2 Fascilitator and Barriers to the Implementation of Evidence into Practice

Several models have been developed to guide healthcare professionals in the successful implementation of evidence into practice.

Selected Models	
Johns Hopkins Model (2005)	A change model which progresses from identification of an EBP question to recruiting and assembling a team; from acquiring, appraising and summarising the evidence to framing practice recommendations; from implementing and evaluating change to communication of findings.
Stetler Model (2001)	A focused model in 5 phases: preparation of research evidence; validation of findings; synthesis of cumulative findings and decision on whether or not to implement a change in practice; translation and practical application of findings; and evaluation as part of routine practice.
Iowa Model (2001)	An organisational model which includes: evaluation of knowledge- and problem-focused triggers; gathering and critique of evidence; decision on whether or not a change in practice is appropriate; and evaluation of structures, processes and outcomes.
Common Elements	
- identify a clinical problem - acquire best evidence - critically appraise the evidence - DECISION: should a change in practice be implemented? - plan and implement practice change - assess outcomes and adjust practice as necessary	

Table 5: Models for Implementation of EBP

17 Adapted from Schub, E., "Evidence-Based Nursing Practice: Implementing," *CINAHL Nursing Guides* (Glendale, CA: CINAHL Information Systems, 2012).

Figure 5: Models for Implementation of Evidence based practice

Facilitators and Barriers to the Implementation of EBP

There are different facilitators and barriers to the Implementation of EBP in institutions. Some of the facilitators and barriers are summarized in table 9

Table 9: Facilitators and Barriers to the Implementation of EBP

Identified Facilitators of the Implementation of EBP	Identified Barriers to the Implementation of EBP
✓ Organizational culture which demonstrates active support for EBP ✓ Provision of EBP education for clinicians and managers ✓ Availability of EBP mentors ✓ Availability of clinical library services ✓ Presence within the organization of EBP champions who will support clinical teams	✓ Absence of organizational support for EBP ✓ Knowledge deficits relating to EBP ✓ Absence of EBP mentors ✓ Negative or apathetic attitude toward EBP ✓ Inadequate access to ICT and/or clinical library services

Session Summary

Identifying Fascilitator and barriers to implementation of evidence into practice is very important for effective implementation of evidence based practice in clinical settings. Furthermore, before applying evidences into practice, recommendations should be evaluated with certain criterias.

References

1. *Leen B, Bell M, McQuillan P. Evidence-based practice: a practice manual.*
2. *Guyyat et al., Users' guide to the medical literature: essentials of evidence based clinical practice . JAMA evidences: 3rd edition*
3. *Busse R, Klazinga N, Panteli D, Quentin W, World Health Organization. Improving healthcare quality in Europe: Characteristics, effectiveness and implementation of different strategies. World Health Organization. Regional Office for Europe; 2019.*
4. *Yentis SM. Decision analysis in anaesthesia: a tool for developing and analysing clinical management plans. Anaesthesia. 2006 Jul;61(7):651-8.*

Chapter Five: Assess Outcomes, Make Changes to Practice, and Quality Assurance

Duration: 4 Hrs

Chapter description: This chapter deals with assessment of changes after applying evidences based practice in clinical settings. The module is delivered in the form of interactive lecture, discussion, computer case scenario-based activities. Trainees' performance will be assessed by self reflection, performance assement (checklist based), review questions and structured posttest.

Primary objective: At the end of this module participants will be able to recognize changes after application of evidence based clinical practice to make sustainable improvements in clinical settings and to apply quality assurance strategies to improve patient outcome.

Enabling objectives: At the end of this session participants will be able to:

- Describe importance of assessing changes after application of evidence based practice
- Identify mechanisms of disseminating findings
- Perform self assement of changes or improvements using the self assement checklist
- Apply quality assurance strategies in EBM

Outline

5.1 Introduction

5.2 How to measure changes

5.3 How to assess your own EBP performance

5.4 Quality Assurance

5.1 Assess Outcomes and Make Changes to Practice

Session Duration: 2 ½ Hr

Specific Objectives: at the end of this session trainees are expected to

Activity 5.1.1: Think-Pair-Share

What are the parameters used to assess changes after application of EBP?

Duration – 15min

Instruction

- T: (Think) about what you know or have learned about the above question- 5minutes .
- P: (Pair) with another participant or a small group and discuss points - 5minutes .
- S: (Share) your thinking with your partner and facilitators expand the "share" into a whole-class discussion- 10 minutes

5.1.1 Introduction

After an evidence-based practice change has been implemented, the final step in the EBP cycle involves assessing outcomes, disseminating results and making further changes to practice as necessary or as prompted by new evidence. Assessing changes after the application of evidence-based clinical decision-making practice is an important part of ensuring that the practice is effective and beneficial for patients. It is important to assess changes in patient outcomes, patient satisfaction, and the overall quality of care provided.

5.1.2 How to measure changes

When assessing changes after the application of evidence-based clinical decision-making practice, it is important to consider the following:

- **Patient Outcomes:** It is important to assess changes in patient outcomes after the application of evidence-based clinical decision-making practice. This can include changes in patient health, quality of life, and satisfaction with care. It is important to measure these outcomes over time to determine if the practice is having a positive impact on patient outcomes.
- **Quality of Care:** It is also important to assess changes in the quality of care provided after the application of evidence-based clinical decision-making practice. This can include changes in the accuracy of diagnosis, the timeliness of care, and the effectiveness of treatments. It is important to measure these changes over time to determine if the practice is having a positive impact on the quality of care provided.

- **Cost effectiveness:** It is also important to assess changes in cost savings after the application of evidence-based clinical decision-making practice. This can include changes in the cost of medications, the cost of tests, and the cost of treatments.

Tools measuring EBP behavior of healthcare professionals should assess the use of EBP steps in practice, the performance of evidence-based clinical procedures and/or the effect of EBP on patient outcomes. All five steps in the EBP process should be considered as part of any assessment.

5.1.3 Disseminating the results of an EBP intervention

Disseminating the results of an EBP intervention may be accomplished in several ways:

Figure 6: Mechanisms of disseminating the results of an EBP intervention

5.1.4 How to assess your own EBP performance

Self-assessment should be an integral part of the continuous cycle of EBP. The checklist below sets out some of the questions you might consider:

Table 10: EBP Self-Assessment Checklist

EBP Self-Assessment Checklist			
EBP step		Yes	No
Step 1	Am I asking any clinical questions?		
Step 1	Am I actively locating evidence/ practice gaps and articulating questions based on same?		
Step 1	Are my questions analyzed into PICO (T) components?		
Step 2	Have I secured immediate access to best evidence via the south East library service?		
Step 2	Do I search the sources of best evidence in my clinical discipline?		
Step 2	Do I use subject headings, limiters and intelligent keywords when searching the main databases: CINAHI, MEDLINE and PsycINFO…?		
Step 2	Do I use the clinical Queries service?		
Step 3	Do I critically appraise the evidence?		
Step 3	Do I integrate critical appraisal measures		
Step 4	Do I adjust critical appraisal measures to the circumstances of my own clinical setting?		
Step 4	Have I implemented an evidence based practice change?		
Step 4	Have I audited the practice change?		
Step 5	Do I continue to assess the quality of patient care and emerging knowledge relevant to the practice change?		

Adopted from: Leen B, Bell M, McQuillan P. Evidence-based practice: a practice manual.

5.2 Quality Assurance

Session Duration: 1 ½ Hr

Session description: this session deals with providing the necessary knowledge on quality assurance measures implemented in evidence-based decision making. The topic covers the important steps of assessing level of quality of care from forming a design team to developing performance indicators..

Specific Objectives: at the end of this session trainees are expected to

- Describe evidence-based medicine quality indicators

- Explain the steps of quality indicator assessment

Activity 5.2.1: Think-Pair-Share

Develop ideas for quality performance indicators to be used in your hospital/work area?

Duration – 15min

Instruction

- T: (Think) about what you know or have learned about the above question- 3 minutes.
- P: (Pair) with another participant or a small group and discuss points – 5 minutes.
- S: (Share) your thinking with your partner and facilitators expand the "share" into a whole-class discussion- 7 minutes

5.2.1 Quality indicators

Quality assurance and quality improvement activities have a complementary and reciprocal relationship with clinical practice guidelines. Quality assurance activities encourage the implementation of guidelines, and guidelines are a crucial component of quality assurance activities

The inclusion of evidence-based guidelines will strengthen formal quality assurance activities such as clinical audit. The information in guidelines may also be used for developing utilization reviews, continuous quality improvement, total quality management and accreditation of institutions or services, as well as providing a reference for interpreting data feedback on clinical units' practices and the health outcomes of their patients

Quality indicators (QIs) are used in assessing the quality of healthcare. Evidence-based clinical practice guidelines (EbCPGs) are relevant sources for generating QIs. In this context, QIs are important tools to assess the implementation of guideline recommendations.

All healthcare personnel should possess the critical competences of EBP and QI (quality implementation). Clinical research, EBP, and QI are distinct but connected fields of study. Systematic, information-driven, change-focused efforts with the goal of enhancing healthcare are what are referred to as QI. Applying evidence-based practice is necessary to have a meaningful impact on improving healthcare quality (EBP). Without EBP, healthcare professionals run the danger of care variations that could have a significant negative impact on patient outcomes.

5.2.1 Quality indicator assessment

In order to identify the level of quality-of-care various types of quality indicator assessments can be used. Most of these assessments follow the following steps

1. **Forming a design team of subject matter experts**

   The measurement team may be stronger if it represents different perspectives. Clinicians can ensure that appropriate clinical indicators are selected together with standards to evaluate whether desirable performance or outcome rates are obtained. Often it may be relevant to select a multidisciplinary team of clinicians, including, for example, doctors from different specialties, nurses, physiotherapists, and occupational therapists.

2. **Developing objectives**

   objectives are defined as the essential things a unit (i.e., a primary care team) does to add value to the organization; in other words, objectives are the main result (and associated characteristics) of the primary care team's work (e.g., amount, quality, timeliness). Examples of health care objectives include patient care performed according to quality standards, effective patient throughput, and personnel allocation matched to patient workload. The key question to answer here is "what is the health care team trying to accomplish when it delivers care?"

3. Generating performance indicators

   A clinical indicator is a measure that assesses a particular health care process or an outcome. In other words, a clinical indicator is a tool for producing a quantitative measurement of quality of care

Examples of clinical process indicators

Indicator concept	Definition of indicator	Type	Standard of care	Time of intervention/treatment	Level of evidence
Organization of treatment (stroke)	Proportion of patients treated/rehabilitated in stroke units	Process	More than 90% of patients with acute stroke should be treated and rehabilitated in a stroke unit	< 24 hours after admission	A
Treatment (lung cancer)	Proportion of patients treated	Process	Treatment rate ≥70%	Discharge	B
Resection (lung cancer)	Proportion of patients resected	Process	Resection rate ≥25%	Discharge	B

A, evidence from at least one randomized controlled trial; B, evidence from quasi-experimental or non-randomized controlled studies.

of car

a) Clinical process indicators. Measuring the quality of a process of care requires determination of whether clinicians are adhering to practices to achieve the best outcomes for patients. generating assessments of value for each indicator

b) Clinical outcome indicators - measurable elements in the outcome of care.

Indicator concept	Indicator definition	Type	Standard of care	Time of intervention/treatment	Level of evidence
Blood glucose control	Proportion of diabetics with HbA1c <7.0 mmol/l	Intermediate outcome	≥90% should have HbA1c <7.0 mmol/l	Every third month	A
Mortality (stroke)	30-day and 3-, 6- and 12-month mortality	Outcome	< 20% should have a 30-day mortality	30 days after stroke	B

A, evidence from at least one randomized controlled trial; B, evidence from quasi-experimental or non-randomized controlled studies.

4. Prioritize indicators by developing contingencies. Contingency development generates a function for each indicator that shows how much the different amounts of the indicator contribute to overall effectiveness. (e.g., the number of days between a fecal occult blood test (FOBT) order and a scheduled colonoscopy signals timeliness in care delivery); (5 vs. 7 days)

Summary

Quality assurance is a process of ensuring that healthcare services meet certain standards of quality, while evidence-based medicine is a practice of using the best available evidence to make decisions about patient care, hence the process involved in developing relevant quality performance indicators (QPI) encompasses the results from evidence-based medicine like clinical guidelines and audit results. These findings are used to develop (QPI) which in turn is used to evaluate and upgrade the quality of health care delivery in a stage wise manner.

Reference

1. *Mainz JA. Developing evidence-based clinical indicators: a state of the art methods primer. International journal for quality in health care. 2003 Dec 1;15(suppl_1):i5-11.*
2. *Hysong SJ, Arredondo K, Hughes AM, Lester HF, Oswald FL, Petersen LA, Woodard L, Post E, DePeralta S, Murphy DR, McKnight J. An evidence-based, structured, expert approach to selecting essential indicators of primary care quality. Plos one. 2022 Jan 18;17(1):e0261263.*

3. *Leen B, Bell M, McQuillan P. Evidence-based practice: a practice manual.*
4. *Guyyat et al., Users' guide to the medical literature: essentials of evidence based clinical practice . JAMA evidences: 3rd edition*
5. *Busse R, Klazinga N, Panteli D, Quentin W, World Health Organization. Improving healthcare quality in Europe: Characteristics, effectiveness and implementation of different strategies. World Health Organization. Regional Office for Europe; 2019.*
6. *Yentis SM. Decision analysis in anaesthesia: a tool for developing and analysing clinical management plans. Anaesthesia. 2006 Jul;61(7):651-8.*

Annexes

Annex 1: EBDM Implementation Checklist

EBP step	EBP Change: Implementation Checklist	Yes	No
Step 1	Ask a clinical question relevant to the patient group , ward or unit		
Step 1	Divide your question into PICO (T) components		
Step 2	Acquire the best evidence from reliable source		
Step 3	Critically appraise and synthesize the evidence		
	Decision: does the evidence imply a change in practice? If Yes, continue. If No, consider another		
Step 3	At this point, it is anticipated that the group are working		
Step 3	Collaboratively with all relevant disciplines		
Step 4	Assess the specific area of clinical practice to get a baseline measure of current status		
Step 4	Agree the scope of practice change to be implemented: how do you envisage the changed practice operating after a month? After 6 months? After 18 months?		
Step 4	Agree a simple audit plan to measure progress, assess patient outcomes and monitor compliance		
Step 4	Agree the duration of a test phase.		
Step 4	As part of the test phase, calculate the costs involved in the proposed change in practice		
Step 4	Agree how the practice change will be communicated to all involved in implementation.		
Step 4	Provide education to healthcare professionals on the rationale for the change in practice and how it will be achieved.		
Step 4	On completion of the test phase, view the results of your audit		
	Decision: do audit results imply a change in practice? if YES, continue. If no, conclude project.		

Step 4	Distribute audit results to ward or unit colleagues.		
Step 4	Amend policies and procedures to account for the practice change		
Step 4	distribute email bulletins, flyers, etc., to publicize the practice change		

Adopted from: Leen B, Bell M, McQuillan P. Evidence-based practice: a practice manual.

References for critical appraisal checklists: *The Pocket Guide to Critical Appraisal; the critical appraisal approach used by the Oxford Centre for Evidence Medicine, checklists of the Dutch Cochrane Centre, BMJ editor's checklists and the checklists of the EPPI Centre.*

Center for Evidence Based Management (July, 2014), Critical Appraisal Checklist for a Cohort or Panel Study. Retrieved (month, day, year) from https://www.cebma.org

Annex 2: Selected Articles for Clinical Appraisal

PLOS | ONE

RESEARCH ARTICLE

Clinical outcome of admitted HIV/AIDS patients in Ethiopian tertiary care settings: A prospective cohort study

Getandale Zeleke Negera*, Teshale Ayele Mega

Department of Clinical Pharmacy, School of Pharmacy, Jimma University, Jimma, Ethiopia

* getandale@gmail.com, getandale.zeleke@ju.edu.et

OPEN ACCESS Freely available online

PLoS one

Prevalence and Determinants of Khat (*Catha edulis*) Chewing among High School Students in Eastern Ethiopia: A Cross-Sectional Study

Ayalu A. Reda[1]*, Asmamaw Moges[1,2], Sibhatu Biadgilign[3], Berhanu Y. Wondmagegn[4]

1 Department of Public Health, College of Health Sciences, Haramaya University, Harar, Ethiopia, **2** School of Public Health, College of Health Sciences, Addis Ababa University, Addis Ababa, Ethiopia, **3** Department of Epidemiology and Biostatistics, College of Medical and Health Sciences, Jimma University, Jimma, Ethiopia, **4** Department of Environmental Health Science, College of Health Sciences, Haramaya University, Harar, Ethiopia

Hindawi
Depression Research and Treatment
Volume 2018, Article ID 1813834, 8 pages
https://doi.org/10.1155/2018/1813834

Hindawi

Review Article

Prevalence and Associated Factors of Perinatal Depression in Ethiopia: A Systematic Review and Meta-Analysis

Amanual Getnet Mersha,[1] Sileshi Ayele Abebe,[1] Lamessa Melese Sori,[2] and Tadesse Melaku Abegaz[3]

[1] *Department of Gynecology and Obstetrics, School of Medicine, College of Medicine and Health Sciences, University of Gondar, P.O. Box 196, Gondar, Ethiopia*
[2] *Department of Psychiatry, College of Medicine and Health Sciences, University of Gondar, P.O. Box 196, Gondar, Ethiopia*
[3] *Department of Clinical Pharmacy, School of Pharmacy, College of Medicine and Health Sciences, University of Gondar, P.O. Box 196, Gondar, Ethiopia*

Journal of Pain Research

Dovepress
open access to scientific and medical research

Open Access Full Text Article

ORIGINAL RESEARCH

Comparing Analgesic Effect of Intravenous Fentanyl, Femoral Nerve Block and Fascia Iliaca Block During Spinal Anesthesia Positioning in Elective Adult Patients Undergoing Femoral Fracture Surgery: a Randomized Controlled Trial

This article was published in the following Dove Press journal:
Journal of Pain Research

Journal of Pharmaceutical Research International

33(50A): 279-286, 2021; Article no.JPRI.76858
ISSN: 2456-9119
(Past name: British Journal of Pharmaceutical Research, Past ISSN: 2231-2919, NLM ID: 101631759)

Comparison of Dexamethasone and Ondansetrone Use in Prevention of Post-operative Nausea and Vomiting

Shakil Alam [a#], Haris Rashid [b], Sagufta Khan [b], Mir Arsalan Ali [b], Obaidul Haseeb [a#] and Santosh Kumar Sidhwani [c‡]

[a] *Ziauddin Hopsital, Pakistan.*
[b] *Department of Surgery, Ziauddin Hospital, Pakistan.*
[c] *Department of Pathology, Ziauddin University, Pakistan.*

Can J Anesth/J Can Anesth (2011) 58:22–37
DOI 10.1007/s12630-010-9407-0

REPORTS OF ORIGINAL INVESTIGATIONS

Perioperative intravenous lidocaine infusion for postoperative pain control: a meta-analysis of randomized controlled trials

Perfusion intraveineuse périopératoire de lidocaïne pour le contrôle de la douleur postopératoire: une méta-analyse d'études randomisées contrôlées

Louise Vigneault, MD · Alexis F. Turgeon, MD · Dany Côté, MD · François Lauzier, MD · Ryan Zarychanski, MD · Lynne Moore, PhD · Lauralyn A. McIntyre, MD · Pierre C. Nicole, MD · Dean A. Fergusson, PhD